

Multifaceted mechanical responsive metamaterials: Mechanisms, fabrications, and applications

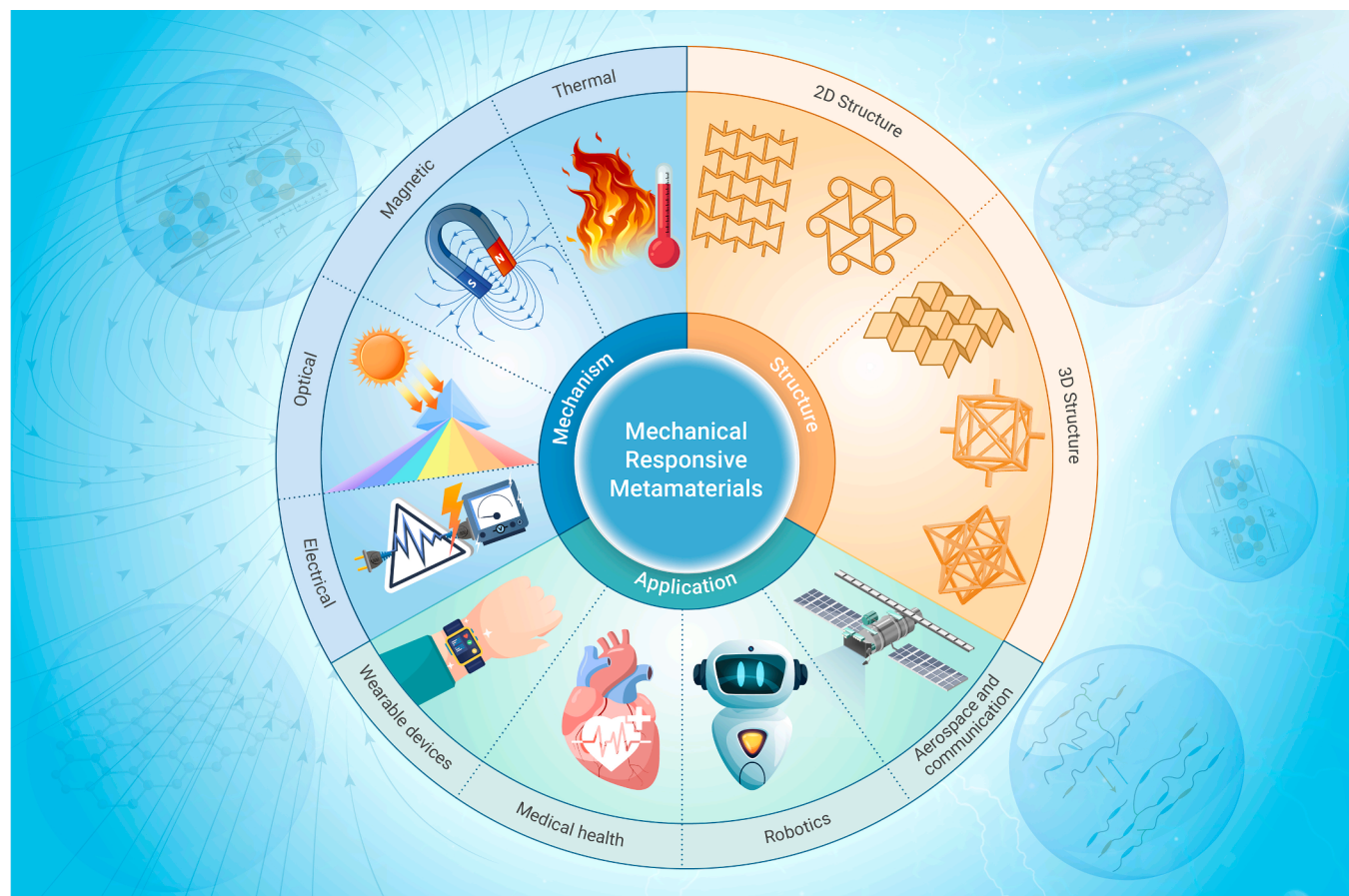
Shiju Yang,^{1,6} Haoshi Ding,^{1,6} Junlong Tan,^{1,6} Yiyao Zhang,^{1,6} Yuqi Guo,¹ Rongjie Kang,¹ Xiaoyang Zhu,^{3,*} Hongzhi Wang,⁴ Ray Baughman,⁵ Zhigang Yin,^{2,*} and Jiuke Mu^{1,*}

*Correspondence: zhuxiaoyang@qut.edu.cn (X.Z.); yinzhg@cqu.edu.cn (Z.Y.); jiukemu@tju.edu.cn (J.M.)

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GRAPHICAL ABSTRACT



PUBLIC SUMMARY

- This review explores mechanical responsive metamaterials that respond to external stimuli such as heat, electricity, and light.
- It highlights the materials and mechanisms enabling stimulus-driven deformations or functional transformations.
- Advanced fabrication methods such as injection molding, 3D printing, and micro/nanoprocessing are discussed.
- Applications across robotics, biomedical devices, adaptive communication, and beyond are examined.

Multifaceted mechanical responsive metamaterials: Mechanisms, fabrications, and applications

Shiju Yang,^{1,6} Haoshi Ding,^{1,6} Junlong Tan,^{1,6} Yiyao Zhang,^{1,6} Yuqi Guo,¹ Rongjie Kang,¹ Xiaoyang Zhu,^{3,*} Hongzhi Wang,⁴ Ray Baughman,⁵ Zhigang Yin,^{2,*} and Jiuke Mu^{1,*}

¹Key Laboratory of Mechanism Theory and Equipment Design of Ministry of Education, School of Mechanical Engineering, Tianjin University, Tianjin 300350, China

²State Key Laboratory of Power Transmission Equipment Technology, School of Electrical Engineering, Chongqing University, Chongqing 400044, China

³Shandong Engineering Research Center for Additive Manufacturing, Qingdao University of Technology, Qingdao 266520, China

⁴College of Materials Science and Engineering, Donghua University, Shanghai 201620, China

⁵Alan G. MacDiarmid NanoTech Institute, University of Texas at Dallas, Richardson, TX 75080, USA

⁶These authors contributed equally

*Correspondence: zhuxiaoyang@qut.edu.cn (X.Z.); yinzhg@cqu.edu.cn (Z.Y.); jiukemu@tju.edu.cn (J.M.)

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Mechanically responsive metamaterials, with their unique structural and functional advantages, have emerged as a key focus in interdisciplinary research spanning materials science, mechanics, and bionics. In this review, we begin with an in-depth analysis of the fundamental materials that constitute mechanically responsive metamaterials, exploring key driving mechanisms such as thermal, electrothermal, electrochemical, photo-responsive, and magnetic responses. By elucidating these underlying principles, we aim to provide readers with a comprehensive understanding of how these responsive behaviors are achieved. The review then systematically evaluates various fabrication methods tailored to specific material properties and structural design requirements, offering a critical analysis of the unique advantages and limitations of each approach. Furthermore, the review highlights the current applications of these fabrication techniques across different fields of mechanical metamaterials, providing a comprehensive overview of their practical implementation and potential.

INTRODUCTION

Metamaterials represent a groundbreaking class of materials that stand at the forefront of innovation in materials science, defining an era where the manipulation of material properties transcends traditional boundaries. Unlike conventional materials, whose properties are dictated primarily by their compositions, metamaterials derive their extraordinary characteristics from their engineered structures.^{1,2} This unique architectural design allows for an unprecedented control over electromagnetic,^{3–7} thermal,^{8–12} acoustic,^{13–16} and mechanical^{17–20} waves, presenting a plethora of advantages across a wide spectrum of industries, including telecommunications,²¹ aerospace,²² healthcare,^{23,24} and energy.^{25,26} The ability to tailor the wave interaction capabilities of metamaterials opens up new avenues for technological advancements that were previously thought unachievable.

Within this broad category, responsive metamaterials emerge as a specialized subset characterized by their dynamic interaction with environmental stimuli. These materials possess the inherent ability to sense changes in their surroundings—such as temperature fluctuations,^{27,28} magnetic field variations,^{29–31} and mechanical forces^{32,33}—and adapt their properties accordingly. This adaptability not only enhances their performance in various applications but also introduces a level of functionality and efficiency that is unparalleled in conventional materials. From thermally adaptive building materials that can regulate indoor temperatures without external energy input to biomedical devices that adjust their behavior based on physiological conditions,³⁴ responsive metamaterials are paving the way for innovative solutions across multiple domains.

The focus of this review is to elucidate the foundational principles that underpin the design and operation of responsive metamaterials. By examining their mechanisms of action, performance advantages, and potential applications, we aim to provide a comprehensive overview of this vibrant field of research. Furthermore, this review navigates through the intricacies of fabrication techniques essential for crafting these advanced materials, delving into methods such as three-dimensional (3D) printing, lithography, and nanofabrication, which

play a pivotal role in achieving the precise geometries and functionalities required by responsive metamaterials.

Covering a broad range of applications, from adaptive architecture and healthcare innovations to telecommunications and aerospace engineering, this review seeks to highlight the transformative impact of responsive metamaterials on modern technology. In doing so, we aspire to offer a detailed insight into the current state of research, the challenges encountered in materials design and application, and the prospective directions that this exciting field may take in the future. Through a meticulous analysis of the principles, mechanisms, and applications of responsive metamaterials, our goal is to underscore their significance in pushing the boundaries of what is possible in materials science and engineering.

ACTUATION MECHANISMS AND SUBSTRATE MATERIALS OF METAMATERIALS

Mechanical responsive metamaterials represent a significant breakthrough in materials science, characterized by their extraordinary ability to adapt and respond to a variety of environmental stimuli. These materials extend beyond traditional passive characteristics, actively altering their properties in reaction to changes in their surroundings. For instance, thermally responsive metamaterials can convert heat into mechanical energy,³⁵ offering revolutionary applications in construction and insulation by adapting to temperature variations. Similarly, when interacting with liquids, these materials can alter their structure,³⁶ leading to innovations in environmental cleanup and advanced filtration technologies.

Expanding their scope, metamaterials that are responsive to electrical and magnetic fields open new avenues in technology and construction. Electrically responsive materials could herald the next generation of smart devices that adjust to electrical changes, while magnetically responsive materials might inspire new medical navigation tools³⁷ and adaptive structural supports.³⁸ Moreover, metamaterials sensitive to light and mechanical forces broaden the spectrum of potential applications for these advanced materials. Light-responsive materials, reacting to different wavelengths, could transform renewable energy³⁹ and medical treatments. In contrast, responsive metamaterials that respond to mechanical forces could play a crucial role in energy harvesting³² from environmental vibrations and in structural health monitoring (Figure 1).³³

The geometric design of these metamaterial structures plays a crucial role in their responsiveness. By meticulously manipulating the micro- to macroscale structures within the material, it is possible to achieve exceptional control over physical phenomena such as light, sound, and heat, as well as a highly sensitive reaction to external stimuli. This design approach not only relies on the physical and chemical properties of the materials themselves but also on advanced mathematical models and computational methods to predict and optimize the performance of metamaterials.⁴⁰

Collectively, these mechanical responsive metamaterials not only mark a significant stride in materials science but also herald a new era of technological innovation. They underscore the immense potential and versatility of these advanced materials, showcasing a wide range of applications from smart textiles and sensors³³ to adaptive structures.^{38,41}

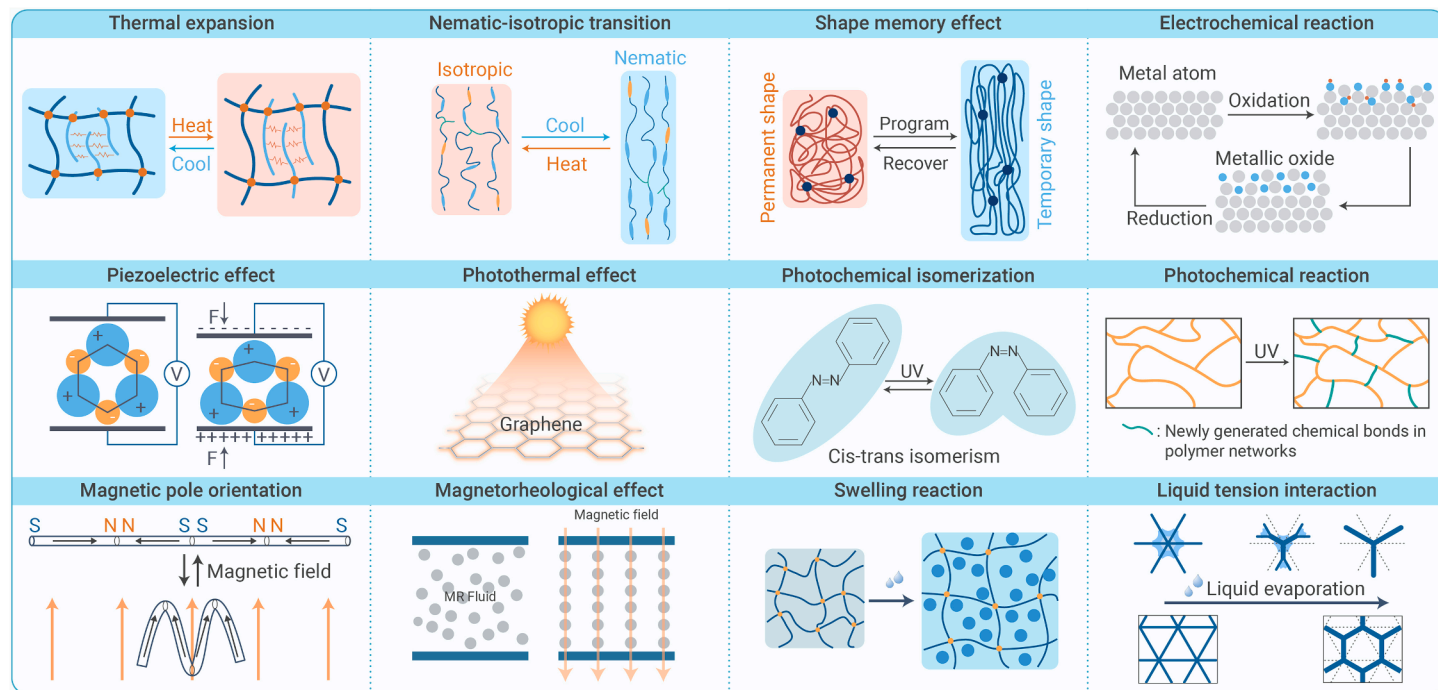


Figure 1. Overview of the response mechanism diagram of responsive metamaterial substrates

In the subsequent sections, we explore the driving mechanisms and substances—the intrinsic materials behind mechanical responsive metamaterials, categorized by their stimuli sources, including thermal, electrical, light, and magnetic influences—while emphasizing the critical role of geometric structure design in their functionality and responsiveness.

Thermally activated mechanisms and substances

To fully grasp the behavior of thermally activated substances in mechanical responsive metamaterials, it is crucial to understand the distinct thermal expansion mechanisms of polymers and metals, and their strategic applications. Polymers undergo significant thermal expansion primarily due to the increased kinetic energy of their molecular chains with rising temperatures. These chains, found in various forms such as linear, branched, or crosslinked, are often randomly arranged in the polymer's amorphous regions. As temperature rises, these chains move more vigorously, causing greater expansion in polymers compared with metals or ceramics. This expansion is further influenced by the polymer's glass transition temperature (T_g), above which the polymer becomes more flexible and expands more noticeably. In contrast, metals exhibit a more uniform and predictable thermal expansion. Their tightly packed atomic structure leads to isotropic expansion—uniform in all directions—as atoms vibrate more intensely with increasing temperature. This expansion is generally less pronounced than in polymers.

Asymmetric thermal expansion. Understanding the principle of asymmetric thermal expansion is essential for the innovative design of thermally mechanical responsive metamaterials. This concept exploits the differential thermal expansion rates of materials to engineer structures that respond predictably to temperature changes, leading to a broad spectrum of applications.

The thermal bimorph effect demonstrates how layers of materials with different coefficient of thermal expansion (CTE), when bonded and heated, bend toward the material with a lower CTE. This phenomenon is instrumental in creating materials with properties such as negative Poisson's ratios and precisely controlled thermal expansion behaviors. Poisson's ratio is a fundamental material property in solid mechanics that describes the ratio of lateral (transverse) strain to axial (longitudinal) strain when a material is subjected to uniaxial loading. It is defined by the following equation

$$\nu = -\frac{\epsilon_{\text{lateral}}}{\epsilon_{\text{axial}}}$$

where ν is Poisson's ratio (dimensionless), $\epsilon_{\text{lateral}}$ is the lateral strain perpendicular to the loading direction, and ϵ_{axial} is the axial strain.

Initial investigations into asymmetric thermal expansion utilized combinations like glass fibers embedded in a polydimethylsiloxane (PDMS) matrix.⁴² This pioneering work laid the groundwork for using thermal expansion disparities to achieve complex, predetermined shapes and functionalities. The patch antenna obtained by combining this deformed structure with liquid metal network has an adjustable resonant frequency, which is more advantageous in dynamic communication compared with traditional patch antennas. Subsequent research expanded the range of materials and techniques, incorporating polyimide (PI) and polymethyl methacrylate (PMMA) using precision laser cutting and rotational casting.⁴³ These advancements allowed for more refined control over material responses to thermal stimuli, opening new avenues in responsive electronics and adaptive optics. In addition to utilizing the difference in CTE between the two materials, it is also possible to control the parameters of the material manufacturing process to achieve changes in the CTE of the same material. For example, by changing the laser power during two-photon lithography (TPL) to alter the thermal expansion rate of the photoresist (Figure 2A),⁴⁴ the thermal bimorph effect can be utilized to construct thermos-mechanical metamaterials. When applied as an actuator in microrobots, this metamaterial enables fundamental motions such as rotation and translation, while offering precise, stable, and reversible responses. In addition, the development of theoretical models has been crucial in designing metamaterials capable of exhibiting tailored thermal expansion behavior.⁴⁵ This approach has facilitated the creation of metamaterials customized for specific thermal-response requirements, enhancing their applicability across various technological domains.

Moving beyond the simple thermal bimorph configurations, the concept of "pivotal junctions" involves the strategic assembly of elements with distinct thermal expansion coefficients at connecting points, resulting in controlled deformation upon temperature changes. This design principle allows for the precise manipulation of a metamaterial's shape and mechanical properties in response to thermal variations. The implementation of copper nanoparticle-doped polyethylene glycol diacrylate (PEGDA) showcased significant advancements, achieving notable negative thermal expansion (NTE).⁴⁶ This NTE behavior results from the combined effects of the mismatch in the CTEs of the constituent materials and the metamaterial's structural design. Due to the lower CTE and higher bulk modulus of copper nanoparticles (CuNPs), CuNP-doped PEGDA exhibits a reduced overall CTE. In the star-shaped structural units of the

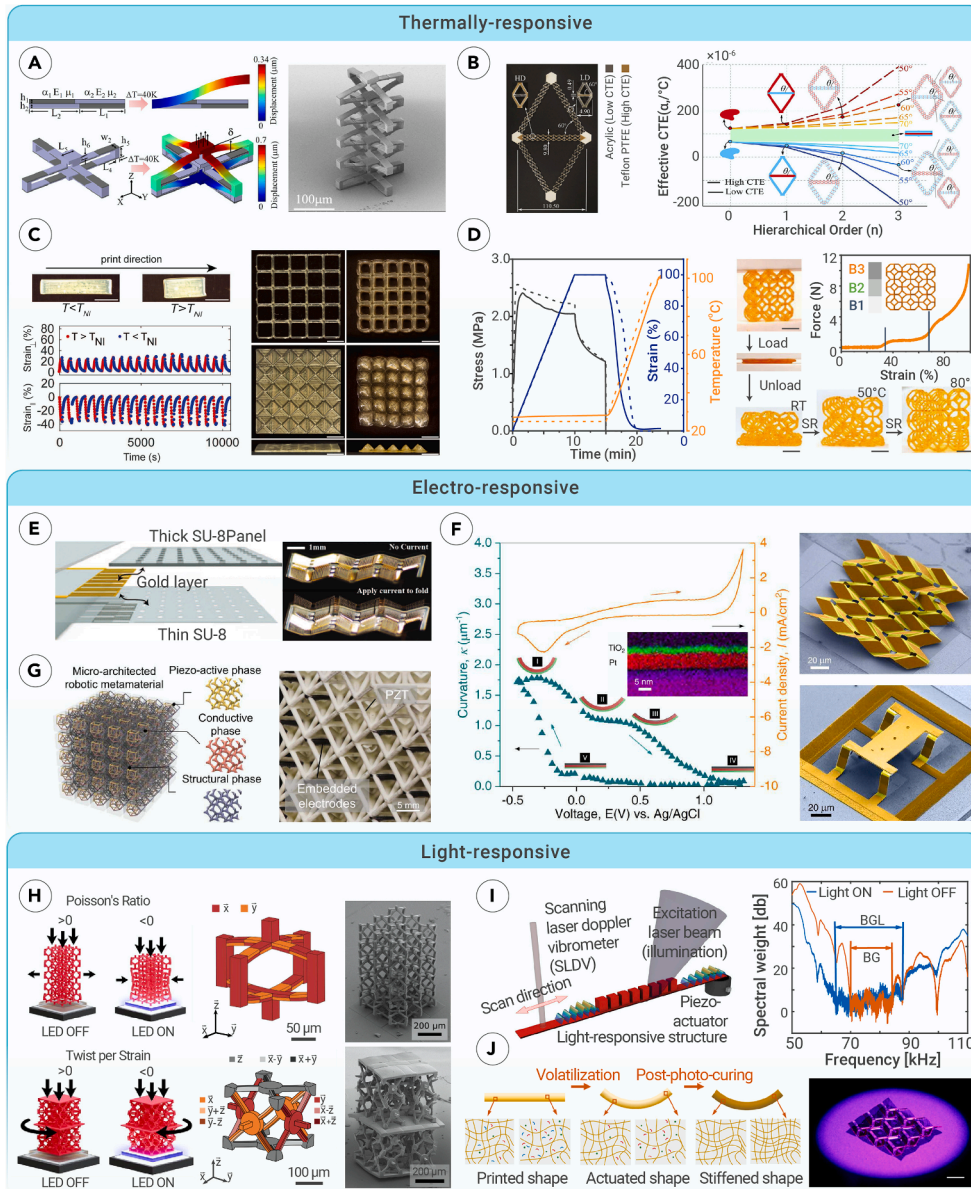


Figure 2. Responsive metamaterials capable of responding to thermal, electrical, and light stimuli (A) Microthermal-responsive metamaterials constructed based on the thermal bimorph effect. Reproduced with permission.⁴⁴ Copyright 2021, The Authors, distributed under CC BY 4.0. (B) Hierarchical metamaterial with almost infinite theoretical CTE constructed based on pivotal junctions. Reproduced with permission.⁴⁸ Copyright 2017, Elsevier. (C) Thermal-responsive metamaterials based on LCE with programmed nematic order. Reproduced with permission.⁵⁰ Copyright 2018, John Wiley & Sons. (D) Shape memory metamaterials that can be cold programmed at room temperature and have multiple triggering temperatures. Reproduced with permission.⁵³ Copyright 2023, The Authors, distributed under CC BY 4.0. (E) Electric responsive micro-origami metamaterial designed based on asymmetric electric thermal expansion structures. Reproduced with permission.⁶⁸ Copyright 2020, John Wiley & Sons. (F) Surface electrochemical reactions actuated micro/nanoscale origami metamaterials. Reproduced with permission.⁷¹ Copyright 2021, The American Association for the Advancement of Science. (G) Integrated piezoelectric metamaterials for actuation and sensing. Reproduced with permission.⁷² Copyright 2022, The American Association for the Advancement of Science. (H) Photothermal-responsive LCE micro/nanochiral metamaterials with electrostatic field-induced orientation. Reproduced with permission.⁹¹ Copyright 2022, Elsevier. (I) Tunable elastic metamaterials activated by photochemical isomerization. Reproduced with permission.⁹³ Copyright 2020, The Authors, distributed under CC BY 4.0. (J) Self-folding metamaterials with stiffness enhancement capability. Reproduced with permission.⁹⁵ Copyright 2020, American Chemical Society.

metamaterial, inner beams of pure PEGDA (high CTE) expand more than outer beams of CuNPs-doped PEGDA (low CTE) during heating. This differential thermal expansion, guided by the structure design, induces coordinated inward twisting or bending, causing macroscopic volume contraction, even though neither constituent material individually exhibits NTE. This research highlighted the potential for adjusting NTE characteristics through material composition and structural design modifications. Further explorations introduced materials such as Al6061 and Ti-6Al-4V into bi-material frameworks,⁴⁷ enabling fine-tuning of the CTE while balancing thermal adaptability with mechanical strength. This innovative approach offered a spectrum of CTE values, from negative to positive, maintaining structural integrity and expanding the potential for aerospace and precision mechanical applications. Based on this concept, the development of graded lattice metamaterials and the integration of polymers such as PTFE and PMMA underscored the possibility of achieving an almost limitless range of CTE values (Figure 2B).⁴⁸ This breakthrough indicated a new era in materials design, where thermal expansion can be meticulously controlled without compromising mechanical properties. The exploration of asymmetric thermal expansion in thermally activated substances represents a critical area of advancement within the field of responsive metamaterials. From the fundamental thermal bimorph effect to the sophisticated pivotal junctions approach, research in this domain has continually pushed the boundaries of what is possible with material

design. Moreover, recent studies on parity-time symmetric thermal diffusion systems have demonstrated an advanced approach for precisely controlling heat flux,⁴⁹ enabling accurate manipulation of both the amplitude and phase of temperature fields. This implies that thermal expansion of materials can also be precisely regulated through such mechanisms, offering a novel insight for the design of thermally responsive metamaterials. The continued development of these materials, characterized by their dynamic responsiveness to thermal stimuli, promises to usher in a new generation of smart systems and devices, transforming our interaction with the material world.

Molecular thermal responses in thermally activated substances. Beyond the principle of asymmetric thermal expansion, molecular orientations and phase transitions also play a pivotal role in enabling macroscopic materials to undergo structural transformations upon reaching specific temperatures.

Liquid crystal elastomers (LCEs) epitomize molecular thermal responsiveness, exhibiting dramatic shape changes when they transition between nematic and isotropic phases at the nematic-isotropic transition temperature (T_{NI}). The thermally induced deformation mechanism of LCEs arises from their unique combination of liquid crystalline order and elastic polymer networks. Upon heating, the aligned mesogenic units within the LCE transition from an ordered nematic phase to a disordered isotropic phase. This phase transition disrupts the anisotropic molecular alignment, resulting in a spontaneous macroscopic contraction along the initial alignment direction of the mesogens. Conversely, upon cooling, the mesogens realign, and the LCE recovers its original shape. This molecular reorientation in LCEs underscores the profound relationship between molecular organization and the macroscopic properties of metamaterials.

The advent of 3D printing with photo-polymerizable LCE inks marked a significant milestone, as liquid crystal molecules achieve orientation during the extrusion process through nozzles, enabling the fabrication of structures with

reversible shape changes under thermal stimuli (Figure 2C).⁵⁰ Under thermal stimulation, LCEs undergo a pronounced contraction along the extrusion direction. By controlling the extrusion printing path, local LCE alignment—and thus thermal deformation direction—can be precisely programmed. This enables reversible 2D-to-3D shape changes, making these high-performance actuating metamaterials highly promising for applications in artificial muscles and soft robotics. Subsequent research expanded these capabilities by adjusting the process parameters of direct ink writing (DIW) 3D printing in real time to alter the local properties of materials, adjusting mechanical stiffness and Poisson's ratio, and broadening LCE applications in responsive devices.⁵¹ In addition, the vertical light attenuation characteristics during digital light processing printing can also be utilized to introduce heterogeneity into the material, just like extrusion processing orients LCEs, enabling rapid production of LCE structures capable of complex movements without physical alignment templates.⁵² Moreover, by introducing light-responsive dynamic bonds, a novel dual-response shape memory effect triggered by light and heat was achieved, enabling the programming and locking of complex actuation shapes on demand, significantly enhancing the versatility of LCE-based responsive metamaterials.⁵³ Recent advancements include the integration of hybrid additive manufacturing for dynamic 3D structures⁵⁴ and novel approaches for biaxial thermal shrinkage,³⁴ pushing forward applications in fields like biomedical engineering. In addition, an innovative study combined LCE materials with different T_{NI} to design a novel class of multi-temperature-responsive LCE metamaterials,⁵⁵ and proposed a method for reverse engineering the deformation behavior of metamaterials, greatly expanding the programmability and deformation mode of metamaterials.

In the evolving landscape of thermally mechanical responsive metamaterials, shape memory polymers (SMPs) emerge as a pivotal category, epitomizing the synergy between molecular reorientation and phase transitions. These polymers are ingeniously designed with a dual-phase structure: a “reversible phase” comprising polymer chains that realign under specific stimuli, and a “fixed phase” of crosslinked segments ensuring structural integrity. This unique dual-phase nature endows SMPs with the remarkable ability to deform in response to temperature fluctuations and external forces, and subsequently revert to their original shape upon exceeding a certain transition temperature. This transformative feature of SMPs allows them to execute a range of mechanical actions, including stretching, bending, and twisting, underscoring their versatility and adaptability in the realm of responsive material design.⁵⁶

The exploration of SMP mechanical responsive metamaterials commenced with the utilization of projection micro stereolithography (PμSL) technology to cure photosensitive precursor solutions of SMPs such as benzyl methacrylate for linear chain construction, alongside poly(ethylene glycol) dimethacrylate, bisphenol A ethoxylate dimethacrylate (BPA), and di(ethylene glycol) dimethacrylate as crosslinkers.⁵⁷ This foundational research achieved micrometer precision in metamaterial structure creation and dynamically adjustable thermomechanical properties, exhibiting highly controllable shape memory behaviors. Precision vascular stents made from such materials can be pre-shaped into a compact form for minimally invasive insertion and activated by heating post-implantation, reducing surgical trauma. In addition, polycaprolactone, a biocompatible and biodegradable SMP, enables airway stents with lower deployment temperatures and improved biosafety.⁵⁸ Further studies introduced formulations involving acrylic acid (AA) and BPA, making AA the backbone of SMP for structural integrity and high T_g , while BPA served as a crosslinker to connect AA chains, forming a network with shape memory characteristics.²⁷ This resulted in significant stiffness changes across a broader temperature range, accommodating applications like tunable damping interfaces and deformable aerospace structures.

Additionally, the emergence of assembly-type SMP materials, such as SMP metamaterials with interface welding achieved through dynamic covalent bond exchange, marked new advancements in customizable module properties.⁵⁹ This result includes the production of 3D Miura pattern structures with zero Poisson's ratio and Kresling pattern cylinders for enhanced mechanical stability, bringing new breakthroughs to the field of shape memory metamaterials. Incorporating Kirigami art into SMP metamaterial design allowed for higher-dimensional thermal-responsive motions and shape changes through precise cutting, folding, and reformation, broadening applications in soft robotics, deployable structures, and adaptive

devices.⁶⁰ SMP-based metamaterials can serve as adaptive torsos and reconfigurable limbs, enhancing mobility over complex terrains and enabling seamless amphibious locomotion.⁶¹

Besides, the creation of multistable mechanical metamaterials using affordable materials like thermoplastic polyurethane (TPU) and shape memory polylactic acid (PLA), characterized by rapid recovery speeds and programmable multistable configurations,⁶² along with cold-programmable shape-shifting metamaterials developed using photocurable resins that deform when programmed at room temperature and recover upon heating (Figure 2D),⁶³ further extended the application range of SMPs in soft robots, deformable antennas, etc. In aerospace applications, the innovative combination of SMPs and origami metamaterials has enabled the development of morphing wing structures,⁶⁴ which greatly enhance the adaptability of aircraft across diverse flight conditions. In the biomedical field, highly stretchable shape memory hydrogels were developed using acrylamide, PEGDA, and water-soluble photoinitiators. Combining it with SMPs based on multi-material 3D printing can produce cardiovascular stents with spatially varying stiffness and drug release functionalities,⁶⁵ offering new possibilities for high-resolution, multi-material 3D-printed structures for specific biomedical applications. Combining SMPs with piezoelectric materials enables bioinspired composites that replicate bone's electromechanical behavior, delivering electrical cues under mechanical strain to stimulate osteogenesis and accelerate bone regeneration.⁶⁶ Recently, research has also integrated low-dimensional nanomaterials such as graphene into SMPs, creating shape memory chiral metamaterials with enhanced mechanical stiffness and thermal conductivity.⁶⁷ These progresses on SMP metamaterials not only highlight innovations in materials science but also demonstrate their profound impact across various technological domains, paving the way for a new era of intelligent adaptive materials and providing transformative solutions across industries. The above exploration of molecular thermal responses in LCEs and SMPs has unveiled a rich landscape of possibilities for designing thermally activated metamaterials.

Electro-responsive mechanisms and substances

Electro-responsive substances form a critical pillar in the advancement of mechanical responsive metamaterials, harnessing electrical energy to induce mechanical changes. This transformation can be achieved through several mechanisms, including electrothermal activation, where electrical currents generate heat to induce material deformation; electrochemical reactions that alter the material's structure or volume; and piezoelectric effects that directly convert electrical energy into mechanical strain. These processes lead to a variety of structural changes in the metamaterials, including bending, rotational movements, and alterations in their modulus.

Electrothermal powered substances. A key aspect of electro-responsive substances is the electrothermal responsiveness. This involves using electric currents to generate localized heating due to the inherent resistance of the material, leading to thermal expansion and softening. Such controlled deformations, like bending or twisting, are executed upon command. Notably, in the electrically induced shape memory effect, metamaterials are pre-set in a temporary shape by heating above their transition temperature. The subsequent application of an electric current, either directly through the polymer or via embedded conductive elements, induces Joule heating. This heating restores mobility to the polymer chains, releasing internal stresses and enabling the metamaterial to revert to its original shape. The recovery process can be designed to be either unidirectional, requiring reprogramming for each cycle, or bidirectional, offering enhanced functionality.

In the field of electrothermal-response mechanical metamaterials, the development of electrically heated micro-origami structures emerged as a pivotal milestone.⁶⁸ This innovation, merging microfabrication techniques with dynamic material responses, facilitated differential heating through gold layer heaters to control shape changes (Figure 2E). This technique introduced new design possibilities for microrobots and dynamic metamaterials, showcasing the complexity of integrating material properties with precision engineering. Further advancements were achieved by embedding a silver nanowire (AgNW) electric heating layer and multi-walled carbon nanotubes (MWCNTs) into a polyurethane acrylate (PUA) matrix, enabling controlled shape transformations under electrical stimulation.⁶⁹ Dynamic mechanical analysis results indicate that doping PUA with MWCNTs enhances its storage modulus, leading

to greater shape recovery stress under thermal stimulation and thus improving shape recovery speed. Additionally, well-dispersed MWCNTs establish strong interfacial interactions with the PUA matrix, which boost mechanical properties such as tensile modulus. However, at higher doping concentrations, MWCNT aggregation occurs, creating microscopic defects that negatively impact both shape memory performance and mechanical strength. Additionally, the incorporation of AgNWs as an electrothermal layer may introduce interfacial defects to some extent, potentially compromising long-term durability. Notably, the application of AgNWs as electrothermal elements and the appropriate doping of MWCNTs resulted in a 25% improvement in the shape recovery speed without the need for increased current. This technology not only enhanced the functionality of shape memory composites but also expanded their applications in flexible actuators and smart devices.⁶⁹ These electrically driven materials demonstrate how localized heating can exploit current to induce controlled material deformation.

Electrochemically powered substances. Electrochemically driven mechanical responsive metamaterials represent a unique and innovative branch within materials science, setting themselves apart from their electrothermal counterparts. Unlike electrothermal metamaterials, which utilize heat generated from electrical currents to drive mechanical changes, electrochemical metamaterials operate through a series of complex electrochemical processes. They include redox reactions, double-layer charge injection, and ion intercalation/extraction, each playing a pivotal role in dictating the materials' mechanical responses. When subjected to an electrical stimulus, these metamaterials experience ion or electron migration, triggering chemical transformations distinct from those in electrothermal materials.²⁵ This process induces mechanical stress, resulting in various deformations such as bending, twisting, expansion, or contraction. This versatility underscores the unique capabilities and potential applications of electrochemically driven metamaterials in the realm of materials science.

The field of electrochemically driven mechanical responsive metamaterials has witnessed continuous advancement with a series of innovative breakthroughs, pushing the envelope for new technologies and laying the groundwork for future applications. A significant starting point in this domain was marked by the development of electrochemically driven micro-actuator technologies.⁷⁰ These actuators, utilizing defined actuation areas in chromium layers and electrochemically grown metal films, along with the volume changes of active polypyrrole under electrical stimuli, achieved precise 3D positioning and reversible movement. The introduction of benzocyclobutene structures to enhance rigidity offered new tools for microfabrication and biological manipulation through a bilayer hinge mechanism. Subsequently, researchers revealed crucial phenomena for the development of lithium-ion batteries by observing the volume expansion and contraction of silicon honeycomb structures during electrochemical lithiation and delithiation processes.²⁵ This discovery underscored the importance of designing electrodes that effectively accommodate volume changes. Furthermore, Greer's team achieved significant morphological changes of silicon microlattice structures triggered by the alloying reaction of silicon with lithium,²⁶ and precise control over the shape of the boundaries of microstructures by implanting artificial defects, opening new avenues for applications in battery electrodes, tunable photonic crystals, and biomedical implants. A pinnacle achievement was the development of microscale, electrically programmable shape memory metamaterials.⁷¹ Specifically, platinum-titanium dioxide thin-film materials fabricated by atomic layer deposition and sputtering can realize fast bending actuation during electrochemical oxidation and reduction reactions on the platinum surface, with rapid actuation times and unique memory functionalities (Figure 2F). This breakthrough introduced a new dimension to microrobots and adaptive structures, showcasing the multifunctionality and transformative potential of electrochemical metamaterials across various applications.

Piezoelectric effect. In the innovative sphere of electromechanical responsive metamaterials, piezoelectric-actuated metamaterials have marked a revolutionary phase, rooted in piezoelectric effects. Generally, piezoelectric-actuated metamaterials harness the piezoelectric effect to dynamically control mechanical properties through electrical stimulation. These metamaterials integrate piezoelectric components within a specially designed structure, allowing for precision in transforming electrical inputs into mechanical responses such as movement or force generation. Their complex architectures are key in optimizing the electromechanical coupling, crucial for applications in adaptive sys-

tems and smart devices. This design approach represents a significant leap in material engineering, offering innovative solutions for advanced robotics, responsive structures, and energy harvesting applications.

In the initial phase of this progression, Cui et al. innovatively incorporated functionalized lead-zirconate-titanate (PZT) nanoparticles with ultraviolet (UV)-sensitive monomers, bonded through a polymer matrix network, creating piezoelectric-actuated metamaterials that demonstrate controlled piezoelectric responses by arranging structural cell patterns spatially within the material, reaching a high piezoelectric charge constant and voltage constant at low volume fractions, and achieving the ability to inversely design an arbitrary piezoelectric tensor.³² It marked a critical step in understanding how spatial arrangement within a material could influence piezoelectric properties, setting the stage for further innovations. On the basis of this work, they further developed a proprioceptive metamaterial-based microrobot through a multi-material 3D printing process,⁷² demonstrating the transition from piezoelectric material development to its application in intelligent systems (Figure 2G). In addition, the innovative combination of acoustic metamaterials and PZT transducers significantly improves the efficiency of ultrasound energy harvesting devices⁷³ and has the potential to be applied to acoustic driving functional devices such as implantable medical devices. However, considering the potential impact of lead-containing piezoelectric materials on the environment and biological health, new high-performance lead-free piezoelectric materials such as barium titanate,⁶⁶ potassium sodium niobate,^{74–76} and Nb-doped $\text{Bi}_{0.5}\text{Na}_{0.5}\text{TiO}_3$ system⁷⁷ have received widespread attention in recent years, which is of great significance for sustainable development.

Light-responsive mechanisms and substances

Light-responsive substances have emerged as a pivotal innovation in the realm of mechanical responsive metamaterials, distinguished by their ability to undergo precise structural and functional transformations upon light exposure. These substances harness the intricate interplay between light and material properties through three primary mechanisms: (1) the photothermal effect, where light is converted to heat causing material deformation, (2) photochemical reactions, such as photochemical isomerization, and (3) photobonding, which induces changes in chemical structure and properties. This multifaceted responsiveness opens up a plethora of applications, ranging from smart windows and actuators to advanced medical therapies and data storage solutions, underscoring the vast potential and versatility of light-mechanical responsive metamaterials in various technological domains.

Photothermal effect. Photothermal materials harness the energy of absorbed light to generate localized heating, leading to material deformation. This mechanism is particularly useful for applications requiring remote actuation or control, as it allows for precise spatial and temporal modulation of material properties.⁷⁸ Gold nanoparticles (AuNPs) with localized surface plasmon resonance (LSPR) are a prime example. LSPR typically occurs in various noble metal nanoparticles and originates from the resonance between incident light and the collective oscillation of conduction electrons on the nanoparticle surface. This resonance is highly dependent on factors such as the size, shape, dielectric environment, and composition of the nanoparticles, leading to strong wavelength selectivity.^{79,80} AuNPs exhibit high photothermal conversion efficiency (PCE) primarily in the near-infrared (NIR) region, typically ranging from 50% to 90%.^{81,82} This efficiency can be further enhanced through optimized synthesis methods. For example, covalently encapsulated gold nanomaterials have demonstrated a PCE of up to 92.8% under 808 nm NIR laser irradiation.⁸³

In contrast, reduced graphene oxide (RGO), as a representative carbon-based photothermal material, exhibits broadband light absorption extending from UV to infrared regions due to its π -conjugated structure and rich π -electron systems. When photons are absorbed, π -electrons are excited from the highest occupied molecular orbital to the lowest unoccupied molecular orbital, followed by vibrational-electronic coupling relaxation processes that efficiently convert photon energy into heat.⁸⁰ Depending on reduction degree, porosity, and functionalization, RGO-based composites typically exhibit PCE values ranging from 40% to 80%,^{84–86} with some aerogel structures reaching over 85% under solar or infrared radiation,^{87,88} making them highly promising for broadband light-harvesting applications.

When these photothermal materials are integrated into metamaterials, their functional performance is influenced by multiple factors, including material

composition and ratio, structural design, and geometric parameters. For instance, a bioinspired deformable hydrogel composed of a poly(*N*-isopropylacrylamide) (PNIPAm) matrix, AuNPs, and fluorinated lithium montmorillonite nanosheets has been developed. Under 520 nm light at 1.06 mW/cm², the photothermal effect of AuNPs rapidly increases the local temperature by approximately 20°C within 3 s, triggering PNIPAm dehydration and shrinkage.⁸⁹ The anisotropic alignment of the nanosheets enables programmable twisting and bending deformations in the metamaterial, with a bending rate of approximately 3°/s. Similarly, RGO-enhanced PNIPAm composite hydrogels can achieve temperature increases of about 30°C within 30 s under broadband light (5 mW/cm², 700–2,900 nm), with a bending deformation rate of around 14°/min.⁹⁰ Furthermore, the introduction of LCE metamaterials incorporating disperse red 1 methacrylate (DR1M) as a photothermal agent has expanded the potential of photothermal-responsive metamaterials. During fabrication, an external electric field is applied to align the liquid crystal domains. The embedded DR1M converts blue light (~450 nm) into heat, resulting in LCE contraction and metamaterial deformation (Figure 2H).⁹¹ Due to the low thermal inertia associated with the micro/nanostructures, these deformations can occur within approximately 0.5 s of illumination.

Photochemical reaction. While the photothermal effect relies on converting light into heat for material transformations, photochemical reactions are fundamentally different. They are triggered by specific wavelengths of light, often UV, that activate internal molecular processes like crosslinking or isomerization, changing the material's physical state from flexible to rigid. An exemplary mechanism is azobenzene's *trans-cis* isomerization, a reversible molecular shift creating significant alterations in material properties such as Young's modulus.⁹² This stark contrast between heat-induced deformation in photothermal materials and molecular-level transformations in photochemical metamaterials highlights their unique capacity to adapt and evolve under light exposure, showcasing the intricate interplay of molecular dynamics within these advanced materials. Such distinct mechanisms underline the versatility of light-responsive substances, enabling a broad spectrum of applications distinct from photothermal metamaterials, ranging from morphing structures to dynamic mechanical systems.

In the field of photochemically driven responsive metamaterials, significant advancements have been made, showcasing the potential of light to control material properties. Initially, researchers developed a photo-responsive elastomer using UV polymerization with materials like bisphenol A ethoxylate diacrylate, incorporating methyl red (MR) as an azo-dopant to enhance its photo-responsivity.⁹³ This material can change its stiffness in response to light due to the photochemical isomerization of MR, making it suitable for applications like dynamic acoustic filtering and wave control (Figure 2I). In addition, reactions such as photobonding can dramatically change material properties; the addition of cellulose nanocrystals (CNC) and cinnamate-modified CNC (cin-CNC) into a polymer matrix enabled the creation of 3D-printed structures with tunable mechanical properties.⁹⁴ This is because the interface between cin-CNC and the polymer matrix can form covalent bonds when exposed to UV light, enhancing the material's adaptability. Another leap forward involved the use of trimethylolpropane triacrylate, aliphatic urethane diacrylate (Ebecryl 8402), and butyl acrylate to develop shape-changing metamaterials.⁹⁵ Using controlled photocrosslinking allowed these materials to undergo rapid and complex shape transformations and possess extraordinary stiffness enhancement ability, indicating the feasibility of dynamic, responsive systems (Figure 2J). The most recent innovation is a photo-responsive hydrogel metamaterial combining polyacrylamide (PAAm), triphenylmethane leucohydroxide, and *o*-nitrobenzaldehyde.⁹⁶ This metamaterial can change its band gap position through patterned UV exposure, offering adjustable functionalities for applications like biomedical devices and underwater acoustics. These advancements illustrate the evolution of photochemically responsive metamaterials, from dynamic stiffness control to programmable shape and structural changes, highlighting their wide-ranging potential for creating complex and adaptable systems.

The exploration of light-responsive substances is rapidly expanding, with ongoing research focusing on enhancing the efficiency, specificity, and reversibility of light-induced responses. By integrating these materials with nanotechnology and bioengineering, the next generation of smart systems and devices will likely exhibit unprecedented levels of adaptability, precision, and functionality. As this field evolves, it promises to unlock novel applications that could

reshape industries, from sustainable energy solutions to advanced healthcare technologies.

Magneto-responsive mechanisms and substances

Magnetic-mechanical responsive metamaterials exploit two primary mechanisms to achieve structural and functional transformations. The strategic manipulation of magnetic poles involves carefully arranged magnetic regions within the material that interact with external magnetic fields, enabling precise and targeted mechanical deformations. This aspect is crucial for applications requiring specific shape or property changes in response to magnetic stimuli. The magnetorheological effect, on the other hand, is observed in fluid mediums. Here, magnetic particles suspended in a carrier fluid align when exposed to a magnetic field, significantly altering the fluid's properties such as viscosity and stiffness.⁹⁷ This change is reversible and offers distinct advantages for dynamic systems, contrasting with the more permanent alterations in solid magnetic metamaterials. The combined use of these mechanisms in magnetic-responsive substances unlocks a wide range of applications, from industrial to biomedical fields, where controlled and reversible transformations are essential.

Recent progress in magneto-mechanical metamaterials marries advanced manufacturing techniques with novel material designs, unlocking transformative applications. Embedding neodymium-iron-boron micro-particles within a silicone rubber matrix has led to materials capable of rapid, reversible shape transformations under magnetic fields, ideal for soft robotics and adaptive devices (Figure 3A).²⁹ The advent of hard-magnetic soft active materials introduces materials that can switch between bending and folding deformation modes in response to magnetic stimuli, with adjustable properties for diverse applications.⁹⁸ Innovations extend to reconfigurable metamaterials constructed from engineered resins and neodymium magnets,⁴¹ acting as dynamic actuators for protective and damping solutions. Further combining with advanced microfabrication technologies, the creation of micro metamaterials composed of modular micro magnetic units that can execute complex behaviors and be reprogrammed *in situ* under the influence of a magnetic field,³¹ demonstrates the broadening scope of these technologies. With the magnetothermal programming metamaterial produced by combining the magnetic torque with shape memory polymers (Figure 3B),⁹⁹ the multistable reprogrammable mechanical metamaterial formed by integrating the bistable structure,¹⁰⁰ and the magnetic-driven Kirigami-metamaterial inspired by Kirigami art,¹⁰¹ these advancements highlight the versatility and potential of magneto-mechanical metamaterials in creating sophisticated, adaptive systems for a range of smart applications. In addition, it is also of great significance for metamaterials based on magnetorheological effects to achieve real-time dynamic control more easily. This type of metamaterial does not require precise control of the distribution of magnetic poles in the material, but rather reserves microchannels for magnetorheological fluids in the metamaterial (Figure 3C), which can adjust the stress-strain characteristics of the metamaterial in real time and remotely as needed through the magnetic field,^{38,102} demonstrating the unique advantages of magnetic-driven metamaterials.

Liquid-mechanical responsive metamaterials

Liquid-mechanical responsive metamaterials, at the forefront of materials science innovation, exemplify the fusion of engineered structures with fluid dynamics to unlock new functionalities and applications. These materials are uniquely capable of adapting their mechanical properties through interactions with liquids, exhibiting phenomena such as swelling, phase transitions, and modifications in surface bonding. The core of their adaptability lies in the materials' design and their physicochemical interactions with solvents, enabling asymmetric swelling, molecular-level reactions (such as pH change), and liquid tension interaction-induced transformations.

A pivotal mechanism in these materials is asymmetric swelling, achieved by utilizing materials with varied swelling coefficients. This leads to specific deformations like bending or stretching upon solvent interaction. A notable application is in hydrogels and other composite materials designed for negative or controlled swelling (Figure 3D).³⁶ Different materials that make up metamaterials experience varying amounts of swelling

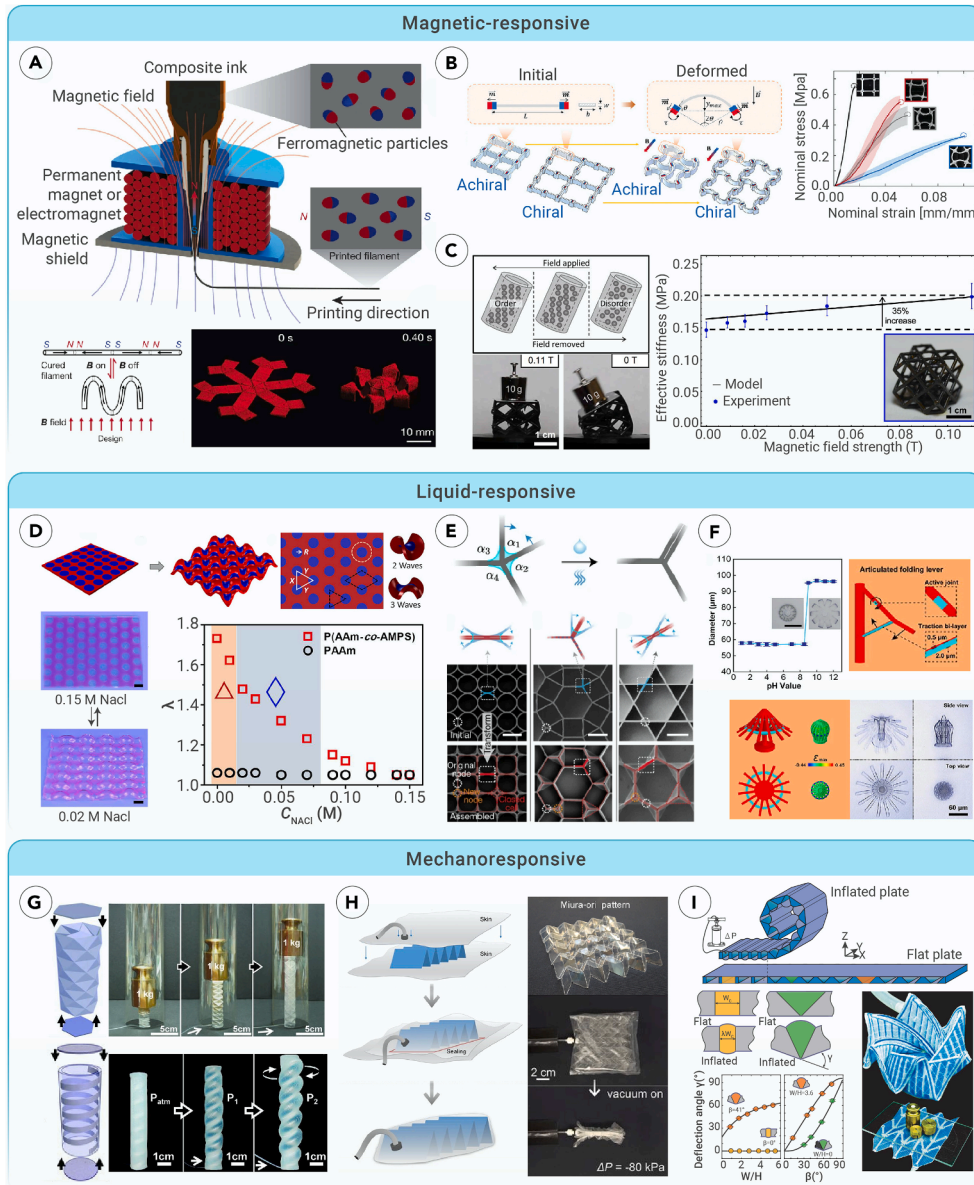


Figure 3. Responsive metamaterials capable of responding to magnetic, liquid, and mechanical stimuli (A) 3D-printed metamaterials with programmed ferromagnetic domains. Reproduced with permission.²⁹ Copyright 2018, Springer Nature. (B) Magnetic thermal-responsive mechanical metamaterials combining magnets with shape memory polymers. Reproduced with permission.⁹⁹ Copyright 2022, John Wiley & Sons. (C) Magnetic field responsive metamaterials based on magnetorheological effects. Reproduced with permission.¹⁰² Copyright 2018, The American Association for the Advancement of Science. (D) Cooperative deformation of patterned hydrogel metamaterials induced by asymmetric swelling. Reproduced with permission.³⁶ Copyright 2017, The American Association for the Advancement of Science. (E) Topology reconfigurable micro metamaterials driven by liquid tension. Reproduced with permission.¹¹¹ Copyright 2021, Springer Nature. (F) Reconfigurable micro metamaterials composed of pH responsive hydrogels. Reproduced with permission.¹⁰⁷ Copyright 2020, The Authors, distributed under CC BY 4.0. (G) Elastomer-cellulose composite origami metamaterials driven by air pressure. Reproduced with permission.¹¹² Copyright 2012, John Wiley & Sons. (H) Origami metamaterials driven by negative pressure. Reproduced with permission.¹¹³ Copyright 2017, The Authors, distributed under CC BY-NC-ND 4.0. (I) Fabric metamaterials with variable Gaussian curvature capability driven by air pressure. Reproduced with permission.¹¹⁷ Copyright 2023, The American Association for the Advancement of Science.

deformation when absorbing solvents, leading to unstable compressive stress within the material and causing buckling deformation to minimize its elastic strain energy. By changing the geometric parameters of the material distribution pattern, this collaborative buckling deformation mode can be programmed. For example, a highly swellable gel sheet constrained by a low-swellable boundary material develops in-plane stresses, leading to out-of-plane bending resembling a sinusoidal profile.¹⁰³ By adjusting the aspect ratio of the unit and the thickness ratio between the two materials, the number of half-waves in the sinusoidal deformation can be effectively tuned. Advanced designs incorporate materials like poly(glycerol sebacate), embedding sodium chloride particles to create microporous structures with rapid volumetric responses to solvent vapors. The material structure with high porosity and low density has more space to accommodate material deformation during swelling, which means less compressive stress will be generated. Therefore, the planar printing structure with gradient porosity along the thickness direction will bend toward the side with low density and low porosity when absorbing solvents. This microporous structure has wide applications in biological scaffolds, biosensors, drug release carriers, and other fields.¹⁰⁴ Further advancements include composite assemblies that combine digital polymers, hydrogels, and rubber elastomers to achieve negative hydration expansion effects. These materials exhibit adjustable responses that can be fine-tuned by altering lattice parameters, opening up applications in bioelectronics and tissue engineering.^{105,106}

parameters (Figure 3F).¹⁰⁷ This has led to the creation of reconfigurable compound metamaterials sensitive to chemical solvents and temperatures, catering especially to fields such as biomedical¹⁰⁸ and information encryption¹⁰⁹ where responsive materials are crucial.

Capillary-induced metamaterials represent a specialized domain within liquid-driven mechanical response metamaterials, driven by capillary forces at liquid interfaces to dynamically deform materials.¹¹⁰ Designed at micro or nano-scales to optimize liquid interactions, they achieve morphological changes such as bending, folding, expansion, or contraction, significantly altering their overall 3D structures. An innovative strategy combines capillary forces with material absorption capabilities to drive complex topological changes in cellular micro-structures, offering rapid, stable, and reversible shape changes that significantly enhance traditional absorption-driven responses (Figure 3E).¹¹¹ Through the interplay of physicochemical properties and fluid environments, these solutions offer new approaches for domains requiring liquid-sensitive functionalities, potentially advancing smart systems and adaptive technologies.

Mechanoresponsive mechanisms and substances

Mechanoresponsive metamaterials, at the forefront of innovation in materials science, are engineered to dynamically alter their mechanical properties, such as shape, stiffness, or vibration behavior, in response to mechanical stimuli. A notable subset of these materials is fluid-driven metamaterials, which specifically respond to changes in pressure exerted by gases or liquids within their

structure. The unique aspect of these materials is their ability to undergo significant transformations due to the pressure variations caused by fluid (gas or liquid) filling or movement.

A key innovation in this field is the development of soft pneumatic actuators capable of anisotropic responses to inflation.¹¹² These actuators leverage materials like Ecoflex and a polyester/cellulose blend (Figure 3G), designed with internal pneumatic networks for controlled expansion and bending. Advancing further, fluid-driven origami-inspired artificial muscles have been introduced, operating under negative pressure to drive motion (Figure 3H).¹¹³ Explorations have also led to materials designed for periodic snap-through and negative stiffness behaviors under compression,¹¹⁴ suggesting new possibilities for energy absorption and impact mitigation. Additionally, soft pneumatic actuators crafted from photo-crosslinked elastomers respond to changes in pneumatic pressure, enabling significant elongation and twisting capabilities, indicative of advancements in soft robotics and artificial muscle technologies.¹¹⁵ Moreover, soft pneumatic linear actuators based on modified triply periodic minimal surfaces represent a leap in design and functionality.¹¹⁶ By channeling air pressure through meticulously designed structures, these actuators achieve uniaxial deformations, underlining the scalability and adaptability of fluidic metamaterials for a broad spectrum of applications. Recently, an innovative study has 3D-printed flexible airbags on fabric surfaces (Figure 3I).¹¹⁷ By designing the cross-section and distribution of airbags, flat fabric sheets can be transformed into various Gaussian curvature surfaces, providing a new approach for the design and manufacturing of large-scale deformable soft robots.

The various response mechanisms of heat, light, electricity, magnetism, etc., mentioned above, correspond to some typical responsive basic materials or specific material combinations. The metamaterials processed from these different basic materials also have some typical structures and different scales. We have summarized this information in Table 1.

FABRICATION METHODS

The development of mechanical responsive metamaterials reflects the integration of innovative design with evolving fabrication techniques. Starting from traditional methods such as casting and machining, the field rapidly advanced with 3D printing technologies such as fused deposition modeling (FDM), stereolithography (SLA), and digital light processing (DLP), enabling the precise construction of complex structures and the integration of diverse materials within a single architecture. Techniques such as DIW and TPL further enhanced material versatility and nanoscale precision, supporting functional integration and fine structural details. Emerging methods, including material jetting (MJ), manual/automated assembly, laser cutting, weaving, etc., have expanded the fabrication toolkit, offering new ways to build dynamic, responsive materials. This evolution highlights that the potential of metamaterials depends not only on theoretical design but also on practical manufacturing.

In this section, we delve into various fabrication techniques employed in the creation of these remarkable materials. From traditional methods that laid the groundwork to contemporary techniques pushing the boundaries, we explore the spectrum of tools and processes that have shaped the metamaterials landscape (Figure 4). Through a meticulous examination, readers will gain insights into how these methods have evolved, the challenges they address, and the opportunities they present for future innovations.

Extrusion-based additive manufacturing technologies

Extrusion-based additive manufacturing technologies play a key role in the world of 3D printing, encompassing several methods such as FDM, melt electro-writing (MEW), and DIW. These technologies share a fundamental process: they extrude material through a nozzle, layer-by-layer, to create intricate objects. This approach is particularly effective for making responsive metamaterials—materials designed to react to changes in their environment, such as temperature shifts or mechanical stress. FDM melts and deposits thermoplastic filaments, making it promising for a wide range of applications, from prototypes to functional parts. MEW, with its precision in drawing ultra-fine fibers, excels at

creating detailed structures on a micro or nanoscale. DIW handles high-viscosity inks to form objects with complex geometries, suitable for materials that respond to light, heat, or magnetic fields. In the coming sections, we explore in detail the three extrusion-based additive manufacturing techniques such as FDM, MEW, and DIW, focusing specifically on their roles in the processing of responsive metamaterials. This discussion will include practical examples that illustrate how each technique is applied in the creation of metamaterials capable of reacting dynamically to external stimuli, thereby pushing the boundaries of innovative material science.

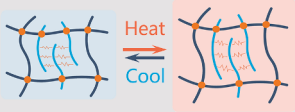
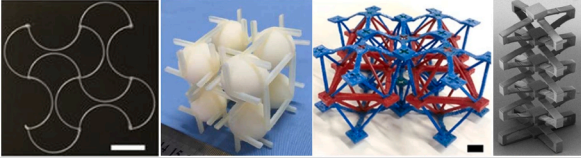
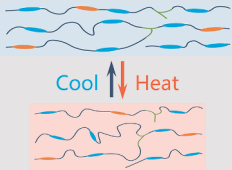
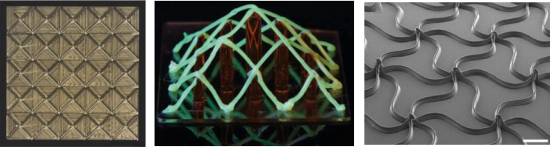
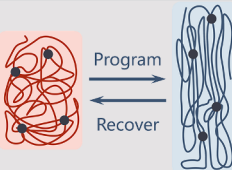
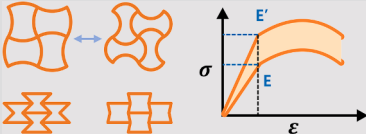
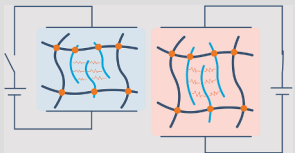
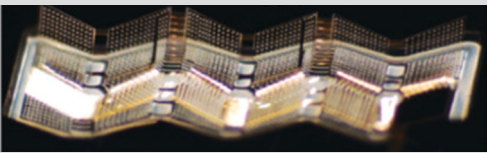
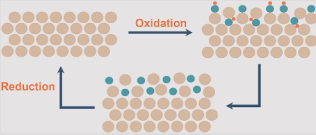
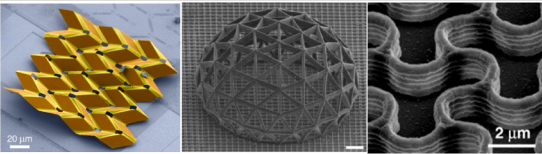
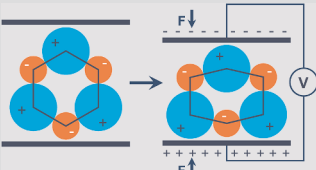
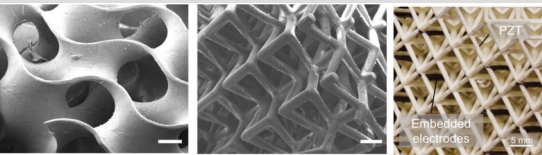
Firstly, FDM emerges as a foundational technology within this domain, leveraging thermoplastic extrusion coupled with computer-aided design to create objects with detailed geometries. FDM's process involves heating and extruding thermoplastic materials such as PLA or acrylonitrile butadiene styrene, layering them to construct 3D form. This versatility is crucial for developing responsive metamaterials, as seen in the work where researchers tapped into the shape memory effects of PLA. Through FDM, they created metamaterials that could undergo significant deformations in response to thermal changes.¹¹⁸ Building on this foundation, the integration of PLA with TPU via FDM led to the development of thermally responsive, reconfigurable metamaterials. These materials are capable of transitioning between multiple states, offering an exemplary case of the technique's potential for complex mechanical responses.¹¹⁹ Furthermore, FDM's adaptation for textile applications—specifically, the creation of active metamaterials featuring patterned TPU airbags on nylon fabrics—opens new avenues for origami metamaterials and shape-shifting robotics.¹¹⁷ An innovative extension of FDM involves incorporating a rotational motion axis to the printing process. This addition allows for the manipulation of thermal deformation modes based on printing path planning, paving the way for applications such as expandable cardiovascular stents,²⁸ demonstrating FDM's adaptability in creating functional structures (Figure 5A). In addition, dual-nozzle FDM printing technology allows for the combination processing of different materials. Combining nylon and polyvinyl alcohol for printing can design metamaterials with customizable negative Poisson's ratio and thermal expansion mode.¹²⁰

MEW, on the other hand, introduces a refined approach to manufacturing by utilizing an electric field to draw polymer fibers from a nozzle onto a substrate. This technique enables the creation of thermally responsive 2D metamaterial lattices with nanometer to micrometer scale fibers (Figure 5C),¹²¹ highlighting MEW's superiority in resolution and its ability to produce ultra-fine structures. Such precision is indispensable for embedding functional elements within metamaterials, marking a significant advancement over traditional FDM.

DIW further expands the capabilities of 3D printing with its precision in depositing high-viscosity inks. This method extrudes viscous fluid ink through a nozzle to construct metamaterials layer-by-layer like FDM, but requires further post-processing to solidify. This processing method greatly expands the range of printable materials, as long as they can be formulated into fluid inks with appropriate viscosity, showcasing its flexibility across a wide range of materials. Among its notable applications is the use of LCE inks to produce structures that can undergo dramatic isotropic transformations when heated, demonstrating DIW's capability to engineer materials with programmable thermal responses.¹²² This feature is further refined by adjusting printing parameters, such as temperature and nozzle diameter, allowing the development of LCE filaments with customizable mechanical properties, including tunable Poisson's ratios.^{50,51} By switching between different printing inks, the DIW method can easily achieve the manufacturing of multi-material structures, utilizing the differences in thermal expansion rates between different materials to create metamaterials with programmable thermal deformation capabilities (Figure 5B).⁴²

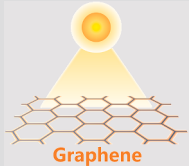
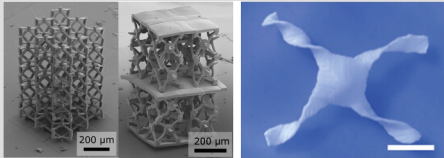
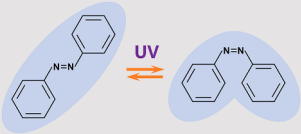
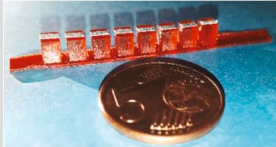
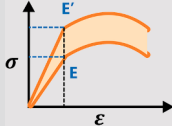
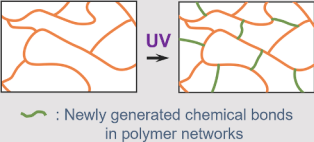
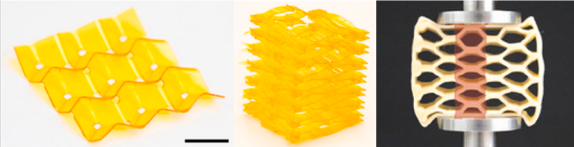
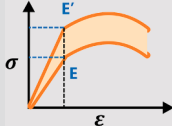
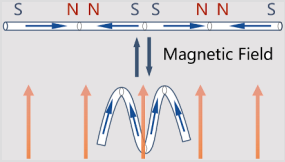
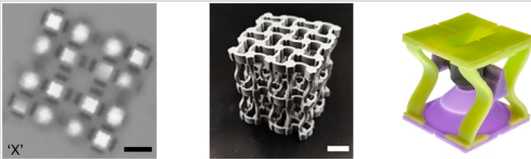
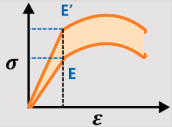
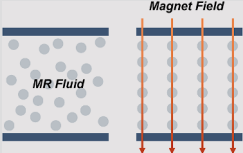
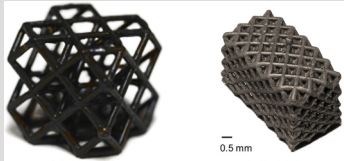
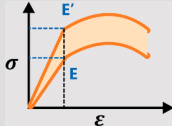
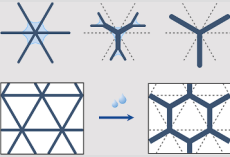
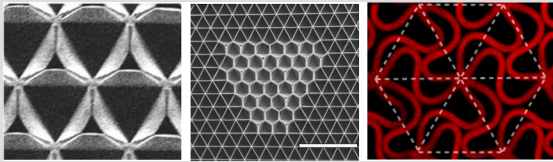
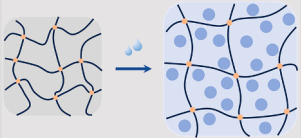
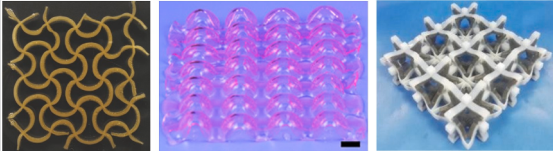
Expanding beyond thermal responsiveness, DIW has facilitated the creation of light-responsive materials using nanocellulose composite inks. When subjected to UV illumination post-printing, these materials exhibit an enhanced structural modulus, illustrating the technique's adaptability.⁹⁴ Moreover, the incorporation of magnetic particles within the ink has opened pathways to developing magnetic-responsive metamaterials, merging materials science with magnetism to manipulate shapes through external magnetic fields.¹²³ The multi-material strategy employed in DIW has proven effective for inducing bending in metamaterials upon heating, showcasing the technique's capability to elicit diverse thermal responses.^{42,124} Addressing the challenges of

Table 1. A summary of typical basic materials, metamaterial structures, and response parameters for various response mechanisms

Response mechanism	Schematic diagram	Responsive materials	Typical metamaterial structure	Variable parameters	References
Thermal expansion		PDMS-glass fibers, PMMA/PI, polyester/PI, nylon/PVA, PE/paper, PTFE/acrylic, SU-8, RGD835/FLX930			Boley et al. ⁴² ; Ji et al. ⁴⁴ ; Xu and Pasini ⁴⁷ ; Wu et al. ¹³⁸
Nematic-isotropic transition		RM82, Irgacure 651, RM257, EDDT, EDT, allyl dithiol, butylated hydroxytoluene, R6M, methyl acrylate, Irgacure 819			Kotikian et al. ⁵⁰ ; Peng et al. ⁵⁴ ; Javadzadeh et al. ¹²¹
Shape memory effect		benzyl methacrylate, PEGDMA, BPA, TPU90A, PLA, TOP31B			Yang et al. ⁶² ; Ge et al. ⁶⁵ ; Wang et al. ⁶⁷
Electrothermal effect		SU-8, PUA, silver nanowire, MWCNT			Zhu et al. ⁶⁸
Electrochemical reaction		silicon, lithium, polypyrrole, gold, Pt, TiO ₂			Baggetto et al. ²⁵ ; Xia et al. ²⁶ ; Liu et al. ⁷¹
Piezoelectric effect		PZT, KNN			Cui et al. ^{32,72}

(Continued on next page)

Table 1. Continued

Response mechanism	Schematic diagram	Responsive materials	Typical metamaterial structure	Variable parameters	References
Photothermal effect		carbon nanomaterial (graphene, RGO, CNT), AuNPs, PNIPAM, polyacrylamide (PAAm), Irgacure 369, DR1M			Zhu et al. ⁸⁹ , Münchinger et al. ⁹¹
Photochemical isomerization		2-(4-dimethylaminophenylazo) benzoic acid			Giozzi et al. ⁹³
Photochemical bonding	 : Newly generated chemical bonds in polymer networks	CNC, BR3741A, vinyl cinnamate, TMPTA, Ebecryl 8402, n-butyl acrylate			Müller et al. ⁹⁴ , Zhang et al. ⁹⁵
Magnetic pole orientation		NdFeB microparticles, Ti, Co, Al		 	Cui et al. ³¹ , Zou et al. ⁹⁹ , Chen et al. ¹⁰⁰
Magnetorheological effect		HDDA, BAPO, MRF-122EG			Zhang et al. ³⁸ , Jackson et al. ¹⁰²
Liquid tension interaction		4''-acryloyloxybutyl 2,5-di(4'-butyloxybenzoyloxy) benzoate)			Li et al. ¹¹¹
Swelling reaction		PEGDA, PEG, BAPO, PAAm, P(Aam-co-AMPS), poly(glycerol sebacate), RGD8530, SUP705, PAAc, ZrOCl ₂			Wang et al. ³⁶ , Liu et al. ¹⁰³ , Wei et al. ¹⁰⁵

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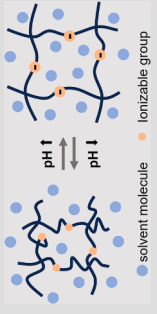
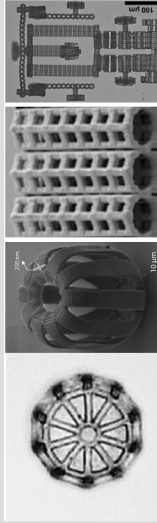
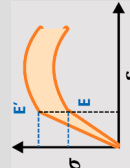
Response mechanism	Schematic diagram	Responsive materials	Typical metamaterial structure	Variable parameters	References
Molecular-level reactions		Aac, NIPAAm, ethyl lactate, PVP, DPEPA, TEA			Jin et al. ¹⁰⁷ , Huang et al. ¹⁰⁸
Deformation:		Modulus change:			Poisson's ratio change:

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constructing complex spatial structures, DIW has been innovatively combined with the DLP technology. This combination allows for the creation of intricate constructs supported by detachable or rigid elements produced through DLP and solves the limitation of traditional DIW that cannot directly print complex suspended structures, marking a significant leap in actuation strain of LCE metamaterials upon heating.¹⁰⁷ Additionally, DIW printing performed within a supporting bath—also known as embedded 3D printing¹²⁵—addresses the challenge of fabricating complex, suspended structures without the need for additional supports.¹²⁶ This approach significantly expands the design freedom and manufacturing capabilities of the DIW technique.

Together, FDM, MEW, and DIW represent a synergistic approach to the development of responsive metamaterials, illustrating the confluence of materials science and engineering in additive manufacturing. By highlighting the unique contributions of each technology, this discussion underscores the potential of extrusion-based manufacturing in pushing the boundaries of material design and responsive system development, paving the way for groundbreaking applications in smart textiles, adaptive infrastructures, and beyond.

Photopolymerization 3D printing

Photopolymerization 3D printing represents a class of manufacturing processes that rely on the precise exposure of light-sensitive resins to controlled light sources. This exposure, characterized by specific wavelengths and intensities, results in the gradual solidification of the resin, allowing for the layer-by-layer construction of intricate objects. Among the prominent techniques in this category are SLA, DLP, and PuSL. These methods have gained widespread use in the production of metamaterials designed for responding to environmental stimuli. In this part of the review, we explore the principles and characteristics of various photopolymerization 3D printing techniques, shedding light on their advantages and limitations in the realm of metamaterial fabrication.

SLA 3D printing, DLP, and g-DLP. SLA and DLP, including its advanced form, grayscale DLP (g-DLP), represent pivotal advancements in the additive manufacturing landscape, each playing a crucial role in the fabrication of mechanically responsive metamaterials. SLA, the earliest invented UV curing technology, utilizes photopolymerization to craft detailed 3D objects through the targeted exposure of liquid photopolymer resin to UV light. A UV laser, guided by mirrors, traces patterns on the resin's surface, initiating a chemical reaction that solidifies the resin point-by-point and layer-by-layer. This method is celebrated for transforming conceptual designs into tangible objects with high detail and complexity, leveraging the accuracy of photopolymerization.

Building on the foundation laid by SLA, the DLP technology accelerates the production process by employing a digital micromirror device (DMD) to project UV light patterns across entire layers of photosensitive resin simultaneously. This shift from the sequential, point-by-point curing of SLA to the rapid, simultaneous layer curing of DLP results in faster production times and parts with smoother surfaces due to uniform exposure. The evolution to g-DLP further enhances DLP's capabilities, introducing the ability to modulate the intensity of UV light across different sections of a single layer. This allows for the creation of components with variable crosslinking densities within the same layer, enabling the strategic alteration of material properties across the printed part. By adjusting the light's intensity, g-DLP can produce parts with graded material properties, significantly broadening the possibilities for designing advanced metamaterials with spatially varied characteristics.

The advancements in DLP and g-DLP, particularly in terms of speed, resolution, and the ability to tailor mechanical properties within a single print session, have opened new avenues for intricate designs and applications in the realm of mechanically responsive metamaterials. These technologies, each building upon the principles of photopolymerization, have transformed the way designers and engineers approach the fabrication of complex structures, pushing the boundaries of metamaterial fabrication and enabling the creation of innovative materials that respond dynamically to environmental stimuli.

One notable application of the DLP technology is the development of patterned hydrogel materials.³⁶ These materials are engineered to swell and transform into predefined shapes upon immersion in water, showcasing good potential for revolutionary applications in soft robotics and optical devices. This transformative behavior exemplifies the potential of DLP in crafting metamaterials that can adapt their shape and function in response to environmental

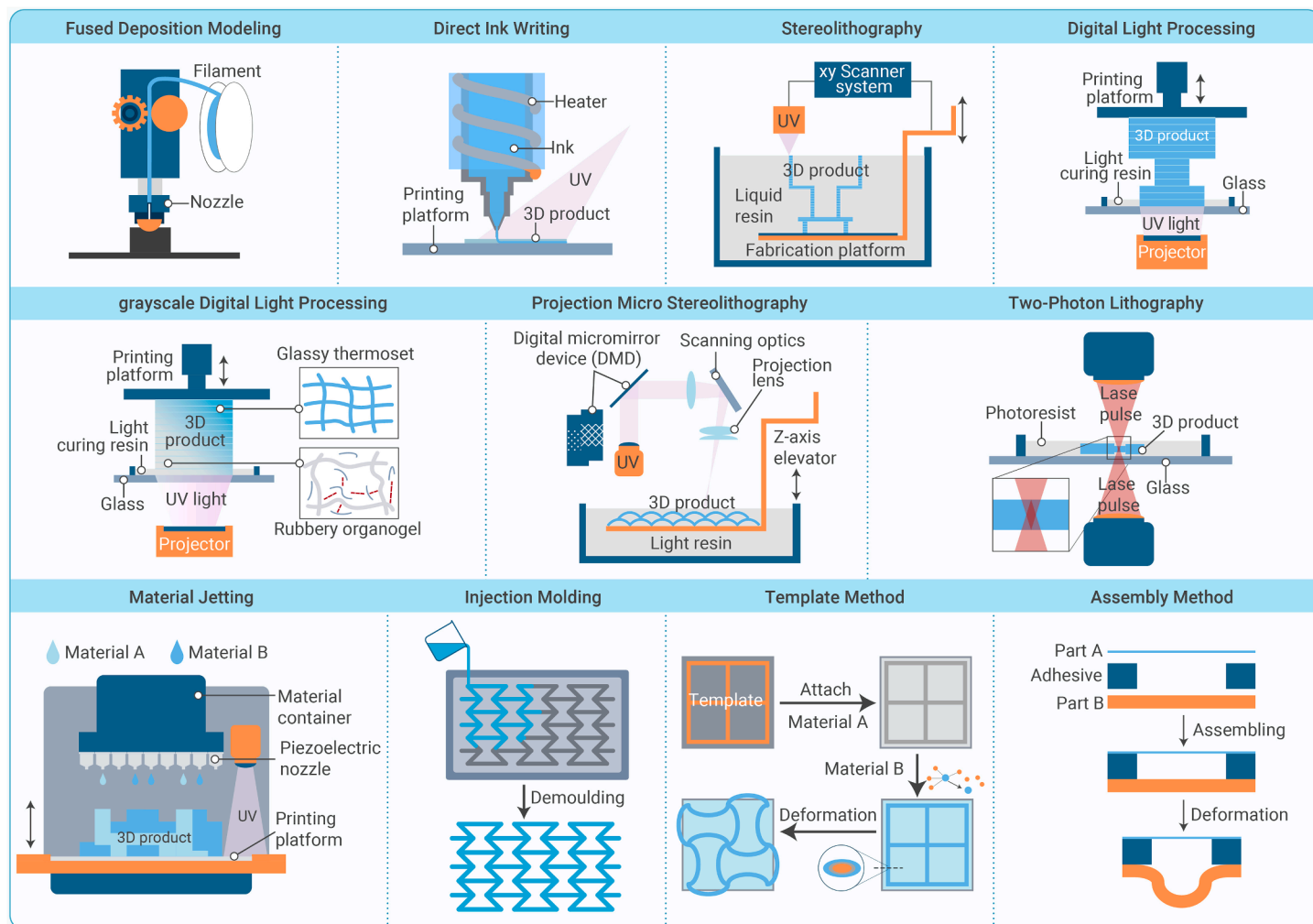


Figure 4. Schematic diagrams of common manufacturing methods for various responsive metamaterials

stimuli. Furthermore, DLP's capabilities extend to the fabrication of 3D metamaterials using SMPs.¹²⁷ These shape memory materials can switch deformation modes when exposed to temperature changes, offering enhanced mechanical performance and adaptive functionality. This application underscores DLP's role in producing materials that not only remember their original shapes but can also transition between different forms based on external temperature stimuli.

Advancements have also been made in developing thermally responsive shape memory metamaterials that exhibit significant strains and recovery ratios.¹²⁸ These materials are envisioned for uses in smart homes and aerospace applications, where their ability to undergo dramatic transformations can be leveraged for adaptive structural and functional purposes. The versatility of DLP is further highlighted in its application to multi-material processing (Figure 5D).¹²⁹ By combining hydrogels, functional polymers, and ceramics, researchers have created hetero-structured metamaterials such as cardiac stents that activate at body temperature and other multi-material prints that integrate different material properties seamlessly.⁶⁵

The advent of g-DLP opens new avenues for materials innovation by allowing light intensity modulation at the pixel level, facilitating the creation of parts with spatially heterogeneous mechanical properties.^{63,95,130,131} This technique has been instrumental in developing materials capable of shape-morphing and pneumatic actuation in response to external forces and thermal inputs, as well as metamaterials that can deform and respond to light to increase modulus after manufacturing (Figure 5E). The ability to engineer materials that vary from highly stretchable gels to rigid thermosetting materials within the same layer illustrates the transformative potential of g-DLP in advancing additive manufacturing. In essence, SLA, DLP, and g-DLP stand as beacons of innovation in the additive manufacturing landscape, heralding a new era of

design possibilities. They exemplify how cutting-edge technologies can transform the conceptual into the tangible, pushing the boundaries of what is achievable in responsive metamaterial fabrication.

P μ SL 3D printing. P μ SL 3D printing is a cutting-edge technique that refines the principles of photopolymerization for the creation of highly detailed and complex 3D structures. This method distinguishes itself from other 3D printing technologies, such as DLP and traditional SLA, through its unique approach to micro projection and scanning construction and its exceptional ability to achieve cross-scale manufacturing from micrometers to centimeters.

The core mechanism of P μ SL involves the use of projection lenses and digital dynamic masks generated by a high-resolution DMD to precisely control the projection of UV light onto a surface of photosensitive resin. Unlike SLA, which scans a UV laser point-by-point across the resin, or DLP, which projects entire layers at once. An objective lens plays a crucial role in P μ SL by focusing the patterned UV light onto the resin surface, significantly miniaturizing the light pattern to achieve feature sizes at the micrometer or even hundreds of nanometers level. This miniaturization allows for the fabrication of parts with intricate details and complex geometries that are not possible with SLA or DLP.

One notable application of P μ SL is the development of materials that exhibit magnetic-mechanical responsiveness. Researchers have utilized P μ SL to construct intricate hollow architectures filled with magnetorheological fluids, which change their modulus in response to magnetic fields,¹⁰² demonstrating the technique's ability to integrate complex structures with functional fluids. In the realm of piezoelectric responsive metamaterials, P μ SL has been pivotal in creating PZT-based piezoelectric nanocomposites. These materials exhibit dynamic voltage responses under stress,³² paving new paths in energy harvesting and sensing technologies through precise material property control. P μ SL has also shown good promise in fabricating temperature-responsive

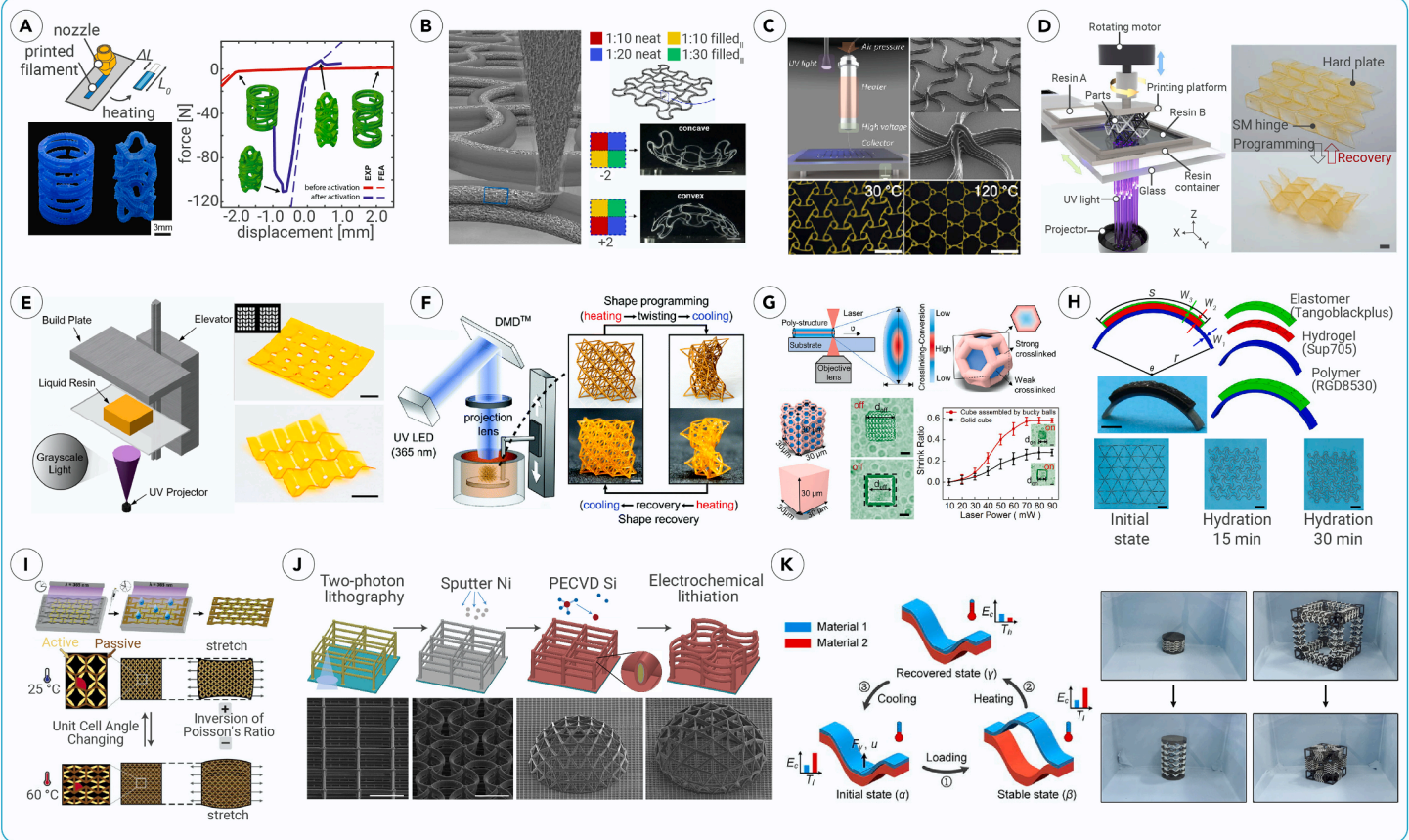


Figure 5. Mechanical responsive metamaterials manufactured using different methods (A) Programmable thermally responsive metamaterials constructed by FDM printing on a rotating axis. Reproduced with permission.²⁸ Copyright 2021, The Authors, distributed under CC BY 4.0. (B) Metamaterials manufactured using the multi-material DIW method. Reproduced with permission.⁴² Copyright 2019, National Academy of Sciences. (C) Micro-nano LCE metamaterials manufactured based on the MEW method. Reproduced with permission.¹²¹ Copyright 2023, John Wiley & Sons. (D) Shape memory metamaterials manufactured based on the multi-material DLP method. Reproduced with permission.¹²⁹ Copyright 2022, The Authors, distributed under CC BY 4.0. (E) Self-folding metamaterials manufactured based on the g-DLP method. Reproduced with permission.⁹⁵ Copyright 2020, American Chemical Society. (F) Shape memory metamaterials manufactured based on the PμSL method. Reproduced with permission.²⁷ Copyright 2019, Royal Society of Chemistry. (G) Light-responsive metamaterials manufactured based on the TPL method. Reproduced with permission.¹³⁷ Copyright 2023, John Wiley & Sons. (H) Negative humidity expansion metamaterial manufactured based on the material jetting 3D printing method. Reproduced with permission.¹³⁹ Copyright 2018, The American Association for the Advancement of Science. (I) Variable Poisson's ratio metamaterials produced by the injection molding method. Reproduced with permission.¹⁴⁴ Copyright 2022, The Authors, distributed under CC BY 4.0. (J) Electrochemical-driven micro metamaterials manufactured based on micro polymer templates. Reproduced with permission.²⁶ Copyright 2019, Springer Nature. (K) Multistable metamaterials constructed by assembling micro units. Reproduced with permission.⁶² Copyright 2023, Elsevier.

metamaterials. By changing the process parameters of the printing process to manipulate the swelling dynamics of PNIPAm hydrogel, researchers have achieved materials capable of gradient response to temperature changes,¹³² showcasing the technique's capacity to tailor responses to thermal stimuli. Additionally, the use of PμSL to produce structures with adjustable stiffness using SMPs (Figure 5F)²⁷ illustrates its ability to create metamaterials that can respond to temperature changes. Moreover, PμSL has enabled the fabrication of polymer lattices with hollow conduits filled with liquid metal gallium (Ga),¹³³ enhancing fracture resilience and introducing shape memory and thermal responsiveness. Through these applications, PμSL has cemented its role as a key player in additive manufacturing, facilitating the creation of advanced metamaterials with precise mechanical, thermal, and electrical responses.

TPL for manufacturing metamaterials. TPL represents a significant leap in the realm of photopolymerization 3D printing, marking a paradigm shift toward precision nanoscale fabrication.^{134,135} This advanced technique diverges fundamentally from the methodologies employed by DLP, SLA, and PμSL. These methods usually cure the topmost photosensitive resin directly by irradiation with UV light, TPL introduces a novel approach by harnessing the simultaneous absorption of two photons from an ultra-short-pulsed laser beam, typically within the NIR spectrum. This interaction occurs at the laser beam's focal point, where the energy of two photons is combined to initiate a photochemical reaction in photosensitive molecules within the resin. The core principle of TPL lies in its use of high numerical aperture optics to achieve an exceptionally tight focus of the laser beam, ensuring that polymerization occurs exclusively at the focal point. This localized interaction allows for the fabrication of complex 3D structures

with resolutions that extend from the submicron down to the nanometer scale, surpassing the capabilities of other photopolymerization techniques. TPL operates on a meticulous layer-by-layer basis. As the focused laser beam methodically scans the resin, it triggers polymerization precisely where the beam is focused, creating a solid structure from the liquid resin. After each layer is completed, the stage holding the resin is lowered incrementally, allowing for the successive addition of layers. This process continues until the desired metamaterial structure is fully realized. Although TPL has extremely high accuracy, its requirements for printing materials and printing environment are very demanding. Usually, printing materials need to be transparent for the working wavelength and need to be manufactured in a yellow light dust-free room, which greatly increases the manufacturing cost of this method.

This advanced technique has opened new possibilities for designing responsive metamaterials with intricate geometries and dynamic functionalities. For instance, TPL has been instrumental in developing materials that exhibit exceptional thermal responsiveness. Researchers have creatively adjusted the laser's writing power across bilayer beam elements, fabricating layers with distinct thermal expansion coefficients.⁴⁴ This innovation has resulted in materials capable of differential bending in response to temperature changes, highlighting TPL's potential for creating thermal switches and adaptive materials. Moreover, TPL's precision and versatility have been showcased in the engineering of "paper-cut" microstructures on pre-stretched LCE films.¹³⁶ These structures undergo reversible deformations driven by temperature-induced strain from the LCE film, paving the way for applications in smart actuators and information encryption technologies. Additionally, the microscale PNIPAm hydrogel grid

metamaterial, prepared using TPL, can deform and reconstruct under thermal excitation, demonstrating entirely different optical and cross-polarization patterns.¹⁰⁹ This phenomenon introduces a novel method for information encryption, further exemplifying TPL's broad application scope.

The technique's capability to finely control material responses has also led to the development of metamaterials with precisely tailored swelling characteristics and deformation responses (Figure 5G).¹³⁷ By exploiting the proximity effect near the focal point to simultaneously achieve weak and strong crosslinking of photosensitive resins, researchers have achieved materials whose deformation responses can be accurately controlled by temperature. This level of control showcases TPL's promise for developing advanced responsive metamaterials and systems.

MJ 3D printing

MJ 3D printing emerges as a sophisticated additive manufacturing method, characterized by its refined technique of depositing liquid resins layer-by-layer to construct intricate objects. This process initiates by warming the resin to a specific temperature range, optimizing its viscosity for accurate deposition. In a method reminiscent of advanced color inkjet printing, the MJ apparatus employs a print head adorned with numerous small nozzles. Governed by piezoelectric modules, each nozzle discharges photopolymer droplets in a precisely coordinated pattern, far surpassing traditional inkjet printers in complexity by building multi-layered structures. Upon deposition, an integrated UV light source instantaneously cures the droplets, effectively transitioning them from liquid to solid forms. This meticulous procedure is repeated, with the build platform descending incrementally after each layer's solidification, culminating in the fabrication of the final 3D artifact.

The multi-material capability of MJ sets it apart, enabling the simultaneous use of diverse photosensitive polymers within a single print job. This feature is pivotal for the creation of responsive metamaterials, offering new possibilities in designing objects with varied properties and functionalities. In exploring the applications of MJ within responsive metamaterials, several key studies illustrate its potential. For instance, the juxtaposition of materials with varying thermal expansion coefficients has yielded anti-chiral structures that exhibit NTE.¹³⁸ Showcasing MJ's capability for creating materials with differential responses to thermal stimuli. Additionally, the integration of high-modulus support materials with low-modulus hydrogels has produced microstructures that bend in response to moisture (Figure 5H),¹³⁹ introducing a novel class of humidity-responsive metamaterials. MJ's precision has also been leveraged to fabricate beam units from hydrogels and RGD8535, which undergo bending deformations under increased humidity, resulting in a negative humidity expansion coefficient.¹⁰⁵ Furthermore, the technology has facilitated the creation of thermal-responsive multistable metamaterials using soft and hard SMPs,¹⁴⁰ capable of switching geometric configurations with temperature variations. Another application saw the development of temperature-responsive metamaterials by controlling the distribution of materials with significant stiffness differences,⁴⁰ enabling varied deformation modes and mechanical properties in response to temperature changes. Recent advancements have extended MJ's applications to processing SMPs into soft grid metamaterials,¹⁴¹ achieving a range of Poisson's ratios and stress-strain characteristics similar to biological skin, promising for artificial skin and thermally activated medical devices. Lastly, the fabrication of defect-insensitive metamaterials with fractal microstructures highlights MJ's role in enhancing material stretchability and resilience,¹⁴² combined with SMPs' thermally triggered shape memory effects, pointing toward potential in wearable technologies and electronic skins. Through these diverse applications, MJ 3D printing underscores its significant contribution to the advancement of responsive metamaterials, showcasing its potential to revolutionize materials science with its unparalleled precision and versatility.

Mold-driven techniques

Injection molding and template method epitomize the innovative strides taken within the field of additive manufacturing, particularly in responsive metamaterials. While these techniques share a foundational reliance on mold-driven processes to shape and functionalize materials, they exhibit distinct characteristics in their approaches to material manipulation and application scope. Injection molding is renowned for its efficiency and precision in mass-producing

polymer-based components. By injecting molten polymer or polymer precursor solution into pre-fabricated molds and allowing it to solidify, this method excels in creating parts with consistent quality and larger feature sizes. Recently, it leverages the advances in 3D printing technologies for mold fabrication, enhancing the ability to produce customized designs. Despite its widespread application, injection molding faces challenges in achieving the microscale to nanoscale details required for intricate metamaterials design, particularly when demolding complex structures. Conversely, the template method offers a tailored approach to metamaterials fabrication by utilizing 3D frameworks, often produced through advanced 3D printing techniques, as scaffolds for the deposition of functional materials. This method enables the strategic integration of various materials, including metals and piezoelectric substances, onto polymer frameworks to impart specific responsive characteristics. Building on the unique advantages and limitations of these two methods, we will next explore their applications in the fabrication of responsive metamaterials separately.

Injection molding. Transitioning from its established roots in conventional manufacturing, injection molding has found a new frontier in the fabrication of responsive metamaterials. Utilizing reversible light-responsive polymers, researchers have developed artificial phototropic metamaterials that mimic the phototropism seen in sunflowers.³⁹ This approach involves injecting a polymer solution into PDMS molds and curing under UV light, leading to structures that bend toward light sources, signaling potential in solar energy and adaptive signal technologies. Expanding the scope of responsive materials, a novel study utilized a two-step injection molding process to create magnetic-mechanical responsive metamaterials.¹⁴³ This method combined silicone gel embedded with magnetic particles to form cilia-like structures, which could generate wave motion under alternating magnetic fields. Such advancements herald new applications in fluid transport and soft robotics. Another research initiative harnessed the contrasting properties of thermoresponsive and non-thermoresponsive hydrogels to engineer metamaterials with tunable Poisson's ratios (Figure 5I).¹⁴⁴ By strategically layering these hydrogels, the resultant metamaterials could dynamically adjust their structural characteristics with temperature changes, demonstrating the adaptability of injection molding in creating temperature-sensitive metamaterials. Further exploration into magnetic responsiveness was conducted by developing silicone elastomer units doped with Fe₃O₄ nanoparticles.¹⁴⁵ Pre-solidification exposure to a magnetic field aligned these particles into directional chains. These units were interconnected using silicone hinges, crafting a magnetic-responsive metamaterial suitable for soft robotics and medical devices. In a distinct application, silicone rubber was employed to design a force-sensing metamaterial.¹⁴⁶ This process involved creating reserved microchannels within the mold, later filled with liquid metal to form a conductive path. External compression altered these channels, offering a unique approach to self-sensing metamaterials by generating signals indicative of the material's stress state.

Combining SMPs with photothermal nanoparticles and PDMS,¹⁴⁷ researchers unveiled light-mechanical responsive metamaterials. Light exposure caused the PDMS to soften, enabling the structure to bend and transition between compression and expansion states, suggesting their applications in adaptive devices. Another team incorporated a low melting point alloy framework within a PDMS film to produce thermally mechanical responsive metamaterials with reversible plasticity.¹⁴⁸ This approach allowed for tunable mechanical properties triggered by heating, illustrating the method's versatility in creating thermally adaptive materials. Lastly, the exploitation of the volume change associated with the gas-liquid phase transition of low boiling point fluids led to the development of a metamaterial with rapid thermal responses.¹⁴⁹ This material expanded upon heating, showcasing injection molding's capability to fabricate materials with tailored responses to thermal stimuli.

Injection molding faces challenges in the realm of responsive metamaterials, particularly in achieving the micro to nanoscale detail required for the most sophisticated designs. Looking ahead, the challenge and opportunity for injection molding could be to overcome its current limitations in feature resolution and demolding intricacy, potentially through the development of new materials, mold design strategies, or post-processing techniques. As researchers continue to push the boundaries of what is possible with injection molding, we can anticipate the emergence of more sophisticated metamaterials with applications

spanning from advanced robotics and adaptive structures to biomedical devices, further solidifying the role of this time-tested manufacturing process in the cutting-edge domain of metamaterials.

Template method. A significant application of template method technique is observed in the development of piezoelectric metamaterials,³³ where polymer scaffold templates created via photopolymerization-based SLA are subjected to selective ion adsorption to deposit copper ions, forming precise sensing electrodes. This process underscores the method's precision in crafting materials for specific uses such as pressure and shock wave detection. Furthermore, resin materials with negative charges have been utilized to generate polymer templates,⁷² directing the deposition of metal ions and polarizing piezoelectric nanoparticles to produce piezoelectric metamaterials. Multiple strain modes can be exhibited under the reverse piezoelectric effect through the internal microarchitecture design of metamaterials, illustrating the template method's versatility.

In another application, polyurethane foam, after undergoing annealing and selective treatments, serves as a template for structures that manifest unique mechanical properties,¹⁵⁰ such as negative Poisson's ratio. A mechanical metamaterial sensor made by using this as a template and adsorbing carbon black has extremely high linearity and sensitivity. This example showcases the method's capacity to produce materials for advanced strain sensors. Additionally, electrochemically responsive metamaterials have been crafted using template deposition techniques,²³ layering nickel and silicon onto polymer lattices (Figure 5J). The expansion of silicon layers caused by lithiation reaction induces structural buckling deformation, highlighting the structural and electrochemical adaptability of the materials produced. Moreover, the template method has enabled the creation of shape memory metamaterials induced by solid-liquid phase transition,¹⁵¹ employing low melting point alloy frames made by injection molding method as templates, suitable for energy absorption and deployable applications. These examples collectively demonstrate the template method's efficacy and adaptability in metamaterials fabrication. It stands out for its ability to engineer materials with tailored functionalities, responding to a wide array of external stimuli, thus marking a significant stride in the advanced manufacturing of responsive metamaterials.

Collectively, injection molding and the template method embody the synergistic essence of mold-driven fabrication approaches within the field of responsive metamaterials. Injection molding is recognized for its durability and scalability, making it optimal for mass production, whereas the template method excels in generating materials with exact functionalities and detailed geometries, perfectly suited for applications requiring extensive customization and precise detailing from the micro to the nanoscale.

Assembly method

The assembly method in the context of metamaterial fabrication stands as a testament to the meticulous craftsmanship required to construct complex structures through the precise amalgamation of various material components. This technique, which encompasses strategies such as manual bonding and thermal compression, finds its niche in crafting structures predominantly at the centimeter scale or larger. It facilitates the creation of intricate 3D constructs by strategically combining simpler modules, showcasing a significant leap toward customizable and complex metamaterial design.

In the domain of thermal responsiveness, the assembly method has led to significant innovations. For instance, the integration of heat-shrinkable polyester with thermally stable PI films exemplifies its capacity to engineer materials that bend and expand upon heating,³⁵ a principle further explored in the construction of multistable metamaterials from PLA, TPU, and PDMS.¹⁵² This type of metamaterial utilizes the stiffness reversal of the two materials during heating to induce the transition of the bistable structure, offering novel solutions for tunable protective gear.

Expanding upon responsive mechanisms, the assembly of bistable structure and shape memory materials showcases the method's ability to combine distinct material modules,⁶² such as buckles and adhesives, and construct large and complex 3D structures. This demonstrates the method's adeptness in material innovation, extending its applications into mechanical memory devices and protective equipment with tunable properties (Figure 5K).

Exploration into magnetic responsiveness has yielded materials capable of transitioning between bistable states under magnetic fields by combining mag-

nets with silicone structures.¹⁰⁰ Further harmonizing materials with contrasting thermal expansion coefficients,¹⁵³ the assembly method facilitates the creation of bistable structures with thermal sensitivity. These innovations open up avenues in soft robotics, mechanical memory devices, and thermosensitive logic devices, underlining the method's adaptability in crafting materials that respond to both thermal and magnetic stimuli. Moreover, the method's exploration into embedding permanent magnets within 3D-printed frameworks⁹⁹ and the design of modular magnetic-responsive metamaterials^{98,154} enriches its repertoire, allowing for programming of uniform magnetization direction and overall structural deformation. This adaptability showcases the assembly method's utility in engineering dynamic structures with unique mechanical, acoustic properties, and potential in electromagnetic wave filtering. Through these thematic clusters of thermal and magnetic responsiveness, the assembly method demonstrates its comprehensive ability to engineer metamaterials across a spectrum of stimuli, consolidating its status as a transformative force in the design and fabrication of responsive metamaterials.

Looking ahead, the assembly method holds promising prospects for further innovation in metamaterial design and functionality. Its ability to integrate diverse materials and achieve precise control over mechanical and physical properties of the resulting structures opens up new possibilities for the development of smart materials and devices. Future research could focus on enhancing the precision and efficiency of the assembly process, exploring new material combinations, and developing modular designs that allow for easy reconfiguration and adaptation to different environmental conditions. As the field of metamaterials continues to evolve, the assembly method will undoubtedly play a pivotal role in unlocking new dimensions of material performance and application.

The various processing methods discussed above have their own characteristics in the manufacturing of responsive mechanical metamaterials. We have summarized the commonly used machinable materials and other characteristics of these processing methods in Table 2.

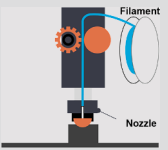
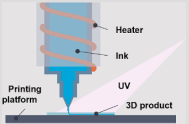
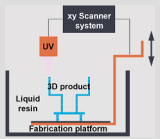
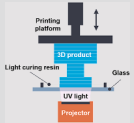
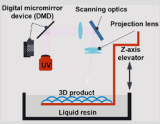
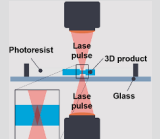
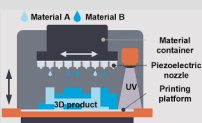
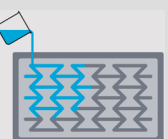
Other techniques

Beyond the above processing methods for mechanical responsive metamaterials, alternative techniques from other fields have also found their place in metamaterial manufacturing, bringing unique approaches to the design and creation of these advanced materials. Techniques such as laser cutting, widely used in flat material processing, and various weaving methods from the textile industry, have been adapted to enhance the fabrication of mechanical responsive metamaterials, showcasing their versatility and innovative potential.

Laser cutting, a method known for its high precision, has been effectively applied in responsive metamaterials fabrication. This technique was utilized to etch precise microgrooves into PI films, which were then filled with a PMMA solution. The distinct thermal expansion coefficients of PI and PMMA facilitated thermally induced shape changes, demonstrating laser cutting's capacity for fine-tuning material properties.⁴³ Further exploration into magnetic-mechanical responsive metamaterials involved using a chromium oxide magnetic PDMS film and NIR illumination to achieve a strategic realignment of magnetic orientations within the material,⁹⁸ showcasing the versatility of laser cutting in programming metamaterial responses. In recent years, femto-second laser processing has emerged as an advanced technique featuring ultrashort pulses and high peak power, enabling precise material removal with minimal thermal damage across a wide range of materials. Its exceptional resolution and nonthermal nature make it ideal for fabricating metamaterials, such as shape memory hydrophobic metasurfaces¹⁵⁵ and thermally responsive metamaterials for droplet manipulation,¹⁵⁶ offering high microscale precision and control.

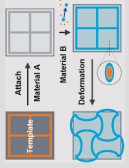
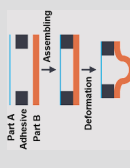
Multi-material multi-nozzle inkjet printing: in the domain of inkjet printing, the conventional paradigm has primarily been dominated by single-nozzle, single-material systems. Such systems, while foundational, inherently operate within a confined design space, primarily due to their material singularity. Their limitation is especially pronounced when addressing the nuanced requirements of metamaterial fabrication, where heterogeneity in material properties often governs the overall performance. Emerging at the forefront of this technological evolution is the multi-material, multi-nozzle printing approach. Unlike its predecessor, this advanced system facilitates the simultaneous extrusion of diverse

Table 2. Comparison of precision, machinable materials, and characteristics of various manufacturing methods

Schematic diagram	Manufacturing method	Resolution	Applicable materials	Advantage	Disadvantage
	FDM	0.2–0.5 mm	various thermoplastic materials, such as PLA, TPU, nylon, etc.	low machine and material costs. Low requirements for printing environment. High technological maturity	poor dimensional accuracy and surface roughness. Printing the hanging structure requires adding support and removing it after printing
	DIW	<1 mm	almost any material is feasible as long as the precursor ink has appropriate rheological behavior, such as PDMS, hydrogel, LCE, etc.	great material adaptability, low equipment requirements and flexible manufacturing	need to repeatedly adjust the formula and rheological properties of ink materials. Post-processing such as solidification is required after processing
	SLA	100–200 μm	liquid photosensitive resins	high applicability of photosensitive resin materials	slow printing speed. The presence of moving parts in the light source results in lower accuracy compared with DLP
	DLP/g-DLP	10–100 μm	liquid photosensitive resins	faster printing speed and higher accuracy compared with SLA	the printing accuracy is limited by the resolution of DMD devices and decreases as the projection area increases
	P μ SL	0.6–10 μm	liquid photosensitive resins	high precision and fast printing speed. Cross scale machining is feasible	the projected pattern has low contrast, requiring precise control of the light source power. The choice of materials is relatively limited
	TPL	<100 nm	transparent liquid photosensitive resins, such as SU8, SCR, etc. (need to be transparent to the working wavelength of the laser)	one of the most accurate light curing printing methods currently available	extremely high requirements for printing environment (yellow light and dust-free room), low printing speed, and only suitable for manufacturing micro and nanostructures
	MJ	50–100 μm	acrylic photosensitive resins, etc.	capable of printing multiple materials simultaneously with high precision. Extremely precise control over the spatial distribution of multiple materials	high requirements for printing material characteristics and high printing equipment and material costs
	injection molding	0.5–1 mm	liquid precursors or melt materials such as silicone elastomer, PDMS and hydrogel	no need for complex equipment, low manufacturing cost, and simple process	difficulty in machining thin-walled and complex spatial structural parts

(Continued on next page)

Table 2. Continued

Schematic diagram	Manufacturing method	Resolution	Applicable materials	Advantage	Disadvantage
	template method	depends on the template	photosensitive resin, metal salt solution, etc.	simple and flexible manufacturing process. Diverse material combinations	the manufacturing process is relatively complex
	assembly method	1–2 mm	various solid materials	complex 3D structures that are difficult to machine directly can be assembled using relatively simple components	the manual assembly process is complex and the accuracy is limited

materials in a single print session. Each nozzle, calibrated for a specific material, orchestrates in tandem with others, enabling the realization of intricate and multifunctional structures. Within the metamaterial context, this capability translates to structures with unprecedented complexity, facilitating the integration of materials with distinct mechanical, thermal, or optical attributes into a coherent entity. Such capability is crucial for integrating materials with diverse mechanical, thermal, or optoelectronic properties into a unified structure.¹⁵⁷

Weaving method: leveraging textile weaving techniques, researchers have introduced a novel approach to metamaterial fabrication by creating thermally responsive structures from woven patterns.¹⁵⁸ By employing highly oriented LCE filaments and integrating them into specific knitted patterns, materials capable of bending deformation in response to temperature changes are produced. This technique highlights the innovative application of traditional weaving methods in producing metamaterials with programmable deformation modes, demonstrating the adaptability and creativity in materials design.

Photolithography process: photolithography is a mature microfabrication technique widely used in the semiconductor industry for its high-resolution, scalable, and batch-manufacturable patterning capabilities. By employing patterned light exposure and selective material removal, it enables precise definition of complex micro/nanoscale geometries over large areas. Recently, it has been applied to metamaterials manufacturing, including micro/nano-metasurfaces^{135,159} and origami-inspired metamaterials,^{31,71,160} leveraging its strengths in precision, repeatability, and scalability. Its integration with emerging materials and design strategies is expected to further advance the scalable fabrication of multifunctional and reconfigurable metamaterials.

These emerging fabrication techniques—including laser cutting, femto-second laser processing, multi-material multi-nozzle inkjet printing, textile weaving, and photolithography—collectively expand the design and manufacturing toolbox for responsive metamaterials. By enabling high precision, multi-material integration, and programmable structural complexity across multiple scales, they open up new possibilities for tailoring metamaterial behaviors to specific functional needs. Their versatility not only enhances fabrication freedom but also accelerates the translation of metamaterial concepts into practical, high-performance applications across microrobots, biomedicine, aerospace, and beyond.

APPLICATIONS OF MECHANICAL RESPONSIVE METAMATERIALS

The high design complexity and unique physicochemical responsive properties of mechanical responsive metamaterials are driving their applications in various fields such as robotics, medicine, acoustics, structural support, aerospace, and digital security. This section explores the diverse applications of mechanical responsive metamaterials, highlighting the groundbreaking advancements they bring to each field and discussing their future potential in shaping the technologies of tomorrow.

Robotic applications

Mechanical responsive metamaterials are revolutionizing robotics by providing versatile power sources and enabling seamless integration of driving and sensing mechanisms. A significant innovation is super-elastic mechanical metamaterials that transition from high stiffness to enhanced flexibility under specific loads.³⁵ This allows robots to adaptively grip various-sized objects without overloading and includes tactile sensing for interactive capabilities (Figure 6A). In addition, metamaterial actuators based on tensegrity structures provide new possibilities for robot design. This type of tensegrity metamaterial can be excited by a magnetic field or motor to undergo programmable linear and bending deformation. Applying it to the legs of a robot can enable the robot to move in any direction (Figure 6B).¹⁶¹ Besides, applying the deformation ability of reversible plastic metamaterials to robots has resulted in reconfigurable robots.¹⁴⁸ By heating with electric current, the structure of the robot is changed from a land robot to a flying robot, greatly improving its environmental adaptability (Figure 6C). Another breakthrough involves a robot metamaterial with bidirectional sensing and driving capabilities, developed through an additive manufacturing technique that merges piezoelectric ceramics with multi-materials.⁷² This enables robots to perform complex movements and respond adaptively to external stimuli, enhancing their functionality (Figure 6D). In terms of artificial muscles used for robots, a fluid-driven muscle inspired by origami

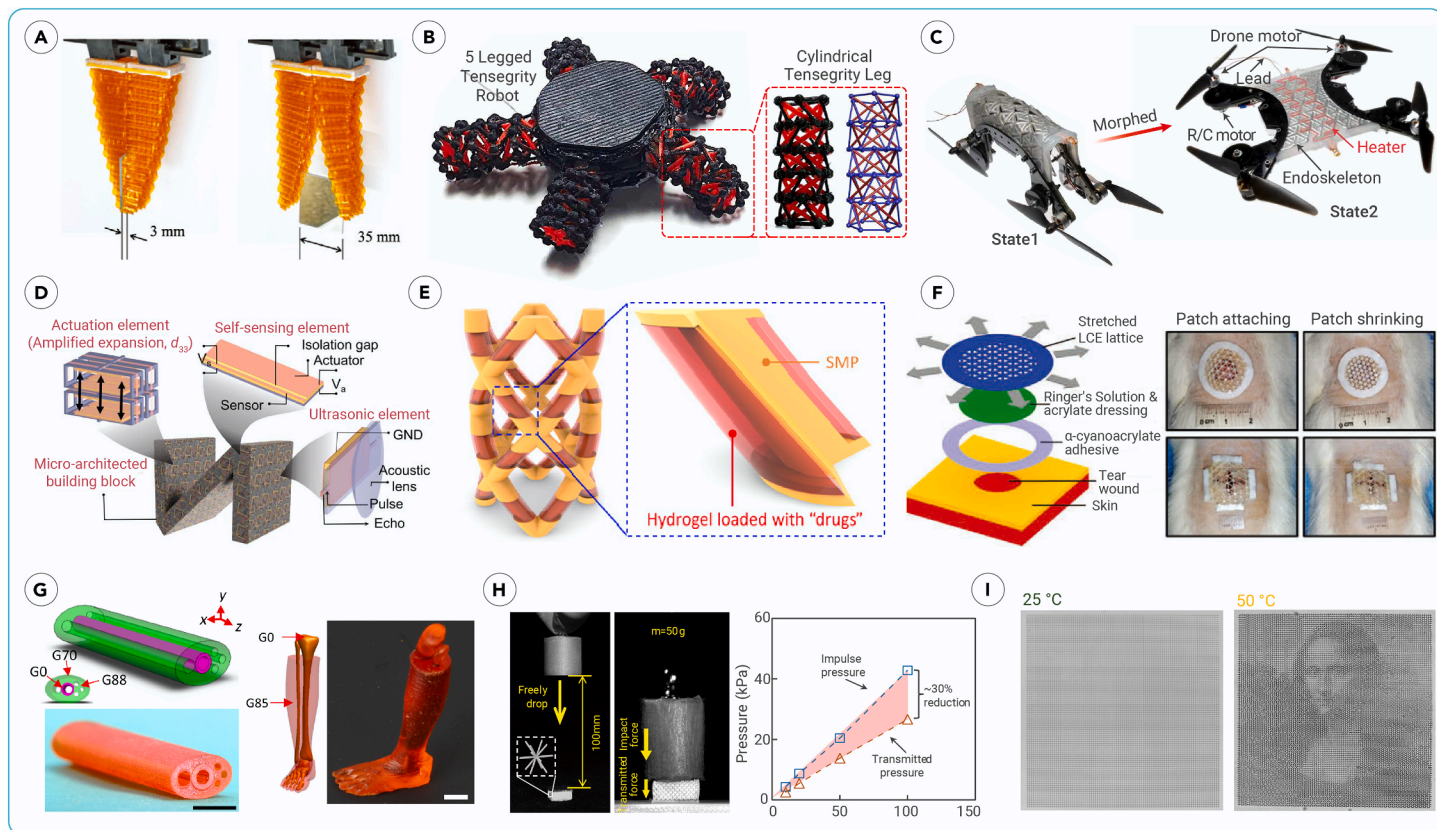


Figure 6. Applications of mechanical responsive metamaterials in various fields (A) Flexible grippers based on super-elastic metamaterials. Reproduced with permission.³⁵ Copyright 2020, John Wiley & Sons. (B) Starfish robots based on tensegrity structures. Reproduced with permission.¹⁶¹ Copyright 2020, The American Association for the Advancement of Science. (C) A variable configuration robot with reversible plasticity. Reproduced with permission.¹⁴⁸ Copyright 2022, The American Association for the Advancement of Science. (D) A self-sensing piezoelectric metamaterial robot. Reproduced with permission.⁷² Copyright 2022, The American Association for the Advancement of Science. (E) Vascular stents made of SMP-hydrogel composite metamaterials. Reproduced with permission.⁶⁵ Copyright 2021, The American Association for the Advancement of Science. (F) LCE metamaterials used to assist in wound recovery. Reproduced with permission.³⁴ Copyright 2021, John Wiley & Sons. (G) Biomimetic metamaterials for human tissue structures. Reproduced with permission.¹⁶² Copyright 2019, The American Association for the Advancement of Science. (H) Piezoelectric metamaterials with impact absorption and self-sensing capabilities. Reproduced with permission.³² Copyright 2019, Springer Nature. (I) Reproduction of encrypted information in hydrogel metamaterials by heating. Reproduced with permission.¹⁰⁹ Copyright 2023, The Authors, distributed under CC BY 4.0.

uses negative pressure for actuation, offering a low-cost, flexible solution for creating linear drives adaptable to various applications.¹¹³ These advancements underscore the pivotal role of responsive metamaterials in pushing the frontiers of robotics, offering innovative solutions that enhance robot functionality, adaptability, and application scope.

Medical applications

In the medical field, the integration of mechanical responsive metamaterials is opening new avenues for treatment, prosthetics, implants, and medical imaging. These materials, particularly those with negative Poisson's ratio characteristics, are proving to be highly effective in applications ranging from drug-delivering vascular stents to wound-healing medical dressings, offering patients more efficient medical solutions. A notable innovation is the design of an SMP-hydrogel composite metamaterial, which can be used as a vascular stent.⁶⁵ This metamaterial can trigger the shape memory effect at 37°C to support blood vessels and release drugs loaded in the hydrogel (Figure 6E). In addition, a noteworthy innovation is the shape memory hydrogel metamaterial based on the phase separation mechanism.¹³⁰ The delayed natural triggering of shape memory effect can be realized by programming the degree of phase separation inside the material. Due to its unique non-heating natural triggering mechanism and the good biocompatibility of hydrogel materials, it has become an ideal material for implants such as vascular stents and probes. Further advancements have been made with a LCE metamaterial that demonstrates exceptional biaxial drive strain and thermal shrinkage properties.³⁴ This metamaterial, when applied to medical dressings, has shown good potential in accelerating skin regeneration and reducing scarring (Figure 6F), based on preliminary rat wound model studies, indicating a significant step forward in non-invasive wound care.

Moreover, the challenge of creating complex artificial tissues and organs has been partially addressed through a novel approach utilizing g-DLP. This technique allows for the printing of structures that mimic the organization, shape, and stiffness of human limbs, including hard bone, soft muscle, and vascular pores (Figure 6G), showcasing the potential for creating more realistic prosthetics and artificial organs.¹⁶² Very recently, the use of DLP and photocurable hydrogel technology¹⁶³ has led to the development of trachea simulation tissues. This 4D bioprinting technique produces scaffolds that mimic biological tissues, demonstrating significant promise for tissue engineering and clinical applications by successfully implanting these structures in damaged rabbit tracheas. In addition to serving as biomimetic tissues, responsive metamaterials can also be used as implants to detect physiological recovery. The combination of metamaterials and triboelectric nanogenerator has resulted in a class of self-sensing metamaterial spinal implants.¹⁶⁴ Implanting this metamaterial into the spinal intervertebral disc will generate electrical signals during spinal activity, and the healing status of the spinal bones can be determined based on the signal pattern and changes. An ultra-thin ultrasound lens with a bacterial cellulose-modified superhydrophobic surface offers significant advances in medical imaging.¹⁶⁵ The micro air chambers on the surface create a strong acoustic impedance contrast with water, allowing precise modulation of the acoustic field via patterned paper-cutting designs. This concentric slit ultrasonic lens enables high-resolution 3D echo imaging, holding great potential for advanced biomedical applications. In addition, ultrasound can provide non-contact stimulation to specific areas of organisms, such as *Caenorhabditis elegans*, after modulation through this type of super lens,¹⁶⁶ which is expected to expand its application in the field of deep brain nerve stimulation. These advancements underscore the transformative potential of mechanical responsive metamaterials in medical applications, from enhancing the efficacy of

treatments to pioneering new methods for tissue engineering and prosthetic development.

Acoustic tunable structures and vibration-absorptive materials

In the realm of acoustics, metamaterials are making significant strides with their unique mechanical structures and tunable properties, offering innovative solutions for elastic wave regulation and impact-resistant material design. A key advancement in this area is the development of globally tunable acoustic band-gap metamaterials, which utilize deformation mode branching to achieve variable acoustic properties.⁹⁸ Its design, driven by magneto-mechanical coupling, allows it to adopt various shapes under different magnetic field orientations, displaying unique acoustic characteristics (Figure 6H). The further integration with shape memory polymer extends its tunability, enhancing its capability for acoustic invisibility. Additionally, a novel electrochemically driven metamaterial has been designed.²³ This metamaterial, transforming a simple tetragonal silicon lattice into a sinusoidal lattice with tunable acoustic properties through electrochemical lithium interaction, opens up opportunities to trap and control phonon modes. Recently, an innovative study proposed a universal design method for high-pixel metasurfaces using angular spectrum propagation and forward optimization,¹⁶⁷ enabling multi-depth, multi-frequency, and complex holography. This ultrasound holography technology holds great promise for medical imaging, underwater detection, and object manipulation. These developments underscore the revolutionary potential of metamaterials in the field of tunable acoustic devices.

Metamaterials are also revolutionizing the field of vibration absorbing materials through their unique tunable properties and innovative designs. These materials offer enhanced carrying capacity and reduced weight due to their regular distribution and hollow structures, significantly improving energy absorption capabilities. The programmable nature of mechanical responsive metamaterials further enriches their functionality in load-bearing and impact resistance applications. Through the use of 3D printing technology, piezoelectric 3D metamaterials have been developed.³² These materials not only absorb over 30% of shock energy but also feature sensing functions that can pinpoint the exact location of impacts. The combination of impact resistance and sensing capabilities in metamaterials introduces new possibilities for smart building applications. Furthermore, the application of shape memory polymers in 4D printing has led to the creation of temperature-responsive metamaterials with adjustable stiffness and reconfigurable shapes.²⁷ Using shape memory effect to change the configuration of metamaterials, the band gaps¹⁶⁸ and Poisson's ratio can be varied significantly. Alternatively, the significant modulus differences of SMPs at different temperatures can be utilized to alter the vibration characteristics of the structure,¹⁶⁹ enabling customizable shock absorption under various load conditions. These applications underscore the transformative impact of metamaterials in providing multifunctional, adaptable, and intelligent solutions for structural support, energy absorption, and smart sensing, marking a significant leap forward in materials engineering and architectural design.

Applications in aerospace and information encryption

In the aerospace industry, the integration of metamaterials is revolutionizing the design and functionality of spacecraft components. The advent of 4D printing has further enhanced the capabilities of shape memory polymer metamaterials, increasing the complexity and functionality of these materials.¹²⁸ This progress allows for the remote adjustment of furniture and the precise deployment of spacecraft solar panels, illustrating the broad applicability of metamaterials in creating smarter living and working spaces.

In the realm of information security, mechanical responsive metamaterials have made significant strides. Micrometer scale paper-cut microstructural metamaterials, produced through two-photon polymerization, hold promise for phonon/photonic devices and secure data encryption.¹³⁶ These intricate designs represent a leap forward in safeguarding information, highlighting the potential of metamaterials in enhancing digital security. Further, the creation of biomimetic stimulus-response metamaterials, inspired by the natural adaptive responses of mimosa plants to environmental stimuli, opens new avenues for all-material intelligent systems. This biomimetic design strategy enables the dynamic programming of materials for intelligent sensing, computing, and communication, marking a significant advancement in the development of

smart devices free from complex hardware dependencies.¹⁷⁰ Additionally, a 2D metamaterial with lattice microstructure capable of topological transformations provides a versatile platform for reversible information encryption and readout. By treating the microstructure with various solvent systems, this technology offers a reliable and adaptable method for information encryption,¹¹¹ further securing the digital landscape. Recently, reconfigurable micro metamaterials based on thermal-responsive hydrogels have provided new possibilities for information encryption. By heating, the synergistic buckling deformation of metamaterial micro units is induced, thereby changing their optical and cross-polarization images (Figure 6I),¹⁰⁹ achieving information display and encryption.

Beyond the applications mentioned, responsive metamaterials offer unique functions such as directional liquid transport control.¹⁵⁵ By creating superhydrophobic microstructures on SMP surfaces, droplet motion can be programmed via surface morphology changes—altering contact angles through pressure and heat stimuli. This reversible and repeatable control benefits fields like chemical engineering and textiles. Additionally, tough hydrogel-based metamaterials serve as protective gear,¹⁷¹ able to repeatedly reshape and conform to objects when incubated in water or precursor solutions, ideal for impact and sports applications.

Looking ahead, the continued development of responsive metamaterials holds great promise for expanding their use across emerging fields. Their programmability, adaptability, and multifunctionality position them as key enablers in next-generation smart materials and devices.

CONCLUSION AND PERSPECTIVE

Mechanically responsive metamaterials integrate diverse material systems and tailored actuation mechanisms—such as thermal expansion, shape memory effects, and anisotropic molecular alignment—to achieve programmable mechanical behaviors under external stimuli. These advanced functionalities are increasingly enabled by additive manufacturing technologies, including FDM, SLA, DIW, and TPL, which offer enhanced structural complexity, material compatibility, and spatial resolution. The synergy between material design, fabrication precision, and structural innovation has opened up broad application opportunities. In the biomedical field, such metamaterials contribute to minimally invasive implants and smart scaffolds; in soft robotics, they enable flexible actuators with programmable motions. Meanwhile, in vibration modulation and acoustic control, they serve as adaptive filters and dampers, and in aerospace and telecommunications, they support deployable structures and tunable antenna systems. These multidisciplinary advances collectively underline the transformative potential of responsive metamaterials in shaping next-generation intelligent systems.

Looking ahead, the future of mechanical responsive metamaterial fabrication is poised for transformative advancements through the integration of both emerging and established techniques. The incorporation of novel processing methods with traditional manufacturing promises to usher in a new generation of materials characterized by unparalleled levels of detail, responsiveness, and multifunctionality. For instance, the integration of artificial intelligence technologies such as machine learning and artificial neural networks into metamaterials design^{172–178} and production¹⁷⁹ processes stands out as a transformative trend. These technologies promise to revolutionize the way we predict and tailor material responses, paving the way for “intelligent” metamaterials with real-time adaptability for applications spanning robotics to biomedical engineering.

Further advancements in multi-material 4D printing promise to extend the functional and structural complexity of metamaterials. This will enable the seamless fusion of diverse materials within a single fabrication cycle, opening up possibilities for novel composite materials and flexible wearable devices with enhanced capabilities. On the nanoscale, advanced techniques such as TPL point to the vast potential of manufacturing materials with intricate nano-architectures. This could lead to the mass production of nanomaterials with tailored properties for advanced sensors,¹⁸⁰ photonic devices, electrical/electronic equipment, and energy solutions. Sustainability and bioinspired designs will also shape the future of metamaterials, focusing on eco-friendly materials^{181,182} and processes. This approach could yield biodegradable, recyclable, or self-healing metamaterials, inspired by nature's efficiency and resilience.

In summary, the integration of emerging and traditional fabrication techniques, combined with the advancements in computational design and

materials science, is poised to redefine the landscape of mechanical responsive metamaterials. This fusion is expected to enhance the functionality and applicability of mechanical responsive metamaterials while also setting the stage for a future where materials can dynamically adapt, offering novel solutions to complex challenges across various sectors.

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AUTHOR CONTRIBUTIONS

S.Y., H.D., J.T., and Y.Z. were responsible for the literature collection, writing of the initial draft, and figure/table preparation. J.M. and Z.Y. supervised the overall structure and focus of the review, provided critical revisions, and are responsible for correspondence during submission and revision stages. R.B., R.K., X.Z., and Y.G. contributed to language polishing, content revision, and final proofreading. All authors contributed to the manuscript and approved the final version.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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